Eidgenössische Technische Hochschule Zürich

Physik III

Autumn Semester 2026

Lecture Notes

Contents

1 Optics **1**
1.1 Fermats Principle . . . 1
1.2 Lenses . . . 2
1.3 Wave Optics . . . 3
1.4 Polarisation of Light . . . 5
1.5 Spectroscopy . . . 7

2 Photons **11**
2.1 Photoelectric effect . . . 11

3 Atoms and their Composition **13**

4 Statistical Mechanics **14**
4.1 The Boltzmann Distribution . . . 14
4.2 System inside a thermal bath . . . 15

5 Wave-Particle Duality **20**
5.1 Localised particles as wave packets . . . 21

6 Quantum Mechanics **22**
6.1 The wavefunction . . . 22
6.2 Physical Variables as Operators . . . 23

References

[1] Home, Jonathan: Vorlesungsskript Physik III. ETH Zürich. 2026

Physik III

Fynn Krebser–fkrebser@student.ethz.ch

Autumn Semester 2026

Lec 1

1 Optics

Optics is the study of light. It is relevant as light is everywhere. Furthermore it is a key tool for measurements in science. Furthermore, it introduces ideas associated with quantum mechanics. Historically, there was a wave and a corpuscle[1] picture of light. This wave/particle duality is the essence of quantum mechanics.

1.1 Fermats Principle

Fermat's principle tries to describe the path of light.

Proposition 1.1: Fermat's Principle

There are two different statements of Fermat's principle.

"Light travels by the route which takes the shortest time."

"Light will take the path which is extremal with respect to variations in the optical path"

The first statement is more intuitive, but the second statement is more general. For example, consider a setup with a mirror

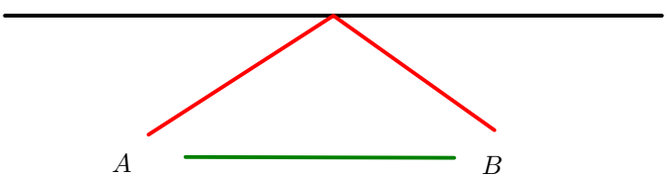


Figure 1: Two people seeing each other with a mirror

Definition 1.2: Optical Path

The OPTICAL PATH is a measure of time taken, written as distance.

$$D = ct = \frac{cs}{v}.$$

Where c is the speed of light and t is the time taken.

Recall that the speed of an electromagnetic wave in a medium is given by $v = \frac{c}{n}$, where n is the refractive index of the medium. Thus we can write the optical path as

$$D = \frac{cs}{v} = ns.$$

[1] Light comes as particles

If the light has a certain frequency f, it will oscillate ever $t_c = \frac{1}{f}$. So the number of cycles in a time t is given by

$$N = \frac{t}{t_c} = \frac{fD}{c} = \frac{D}{\lambda_0},$$

where λ_0 is the FREE-SPACE WAVELENGTH of light.

The optical path tells us how many cycles of oscillation we have convered on our path.

We now want to discover the refraction law using fermat's principle.

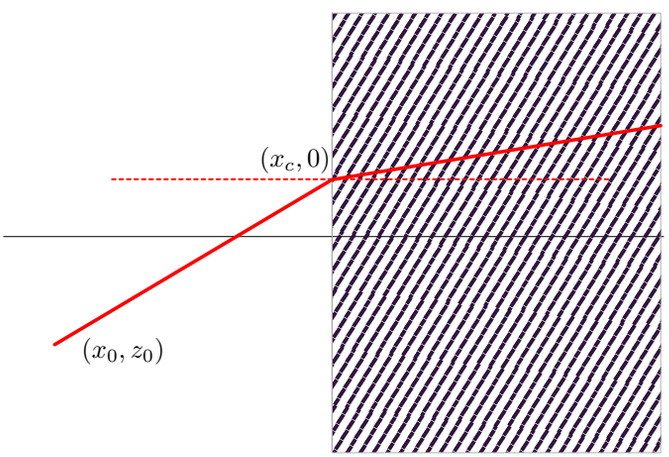


Figure 2: Deriviation of Snell's law using Fermat's principle

The optical path in the left half is given by

$$D_1 = n_1\sqrt{(x_c - x_0)^2 + z_0^2}).$$

In the right half, we have

$$D_2 = n_2\sqrt{(x_p - x_c)^2 + z_p^2}).$$

So we want to minimize the total optical path $D = D_1 + D_2$ with respect to x_c. We have

$$\frac{dD}{dx_c} = \frac{dD_1}{dx_c} + \frac{dD_2}{dx_c} = 0$$

$$\frac{n_1}{\sqrt{(x_c - x_0)^2 + z_0^2}}(x_c - x_0) = \frac{n_2}{\sqrt{(x_p - x_c)^2 + z_p^2}}(x_p - x_c)$$

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2).$$

Where θ_1 and θ_2 are the angles of incidence and refraction, respectively. This is known as Snell's law.

Theorem 1.3: **Snell's Law**

A light ray incident on a boundary between two media with refractive indices n_1 and n_2 will be refracted according to the law

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2).$$

1.2 Lenses

The best way to think of a lens is to think about the eye. It consists in principle of a lens and a screen (the retina). In principle, the lens is of refractive index n_2 in some sort of substance of refractive index n_1. We are restricted by the shape and the refractive indices of the lens. Furthermore, we are restricted by the size of the lens.

Let us focus on the ideal lens.

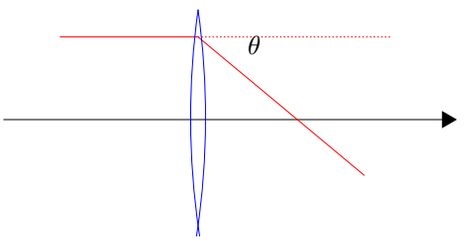


Figure 3: Ray bending of an ideal lens.

Light hitting the lens at a height h is deflected by the angle $\theta = \frac{h}{f}$, where f is the FOCAL LENGTH of the lens. Light incoming parallel to the optical axiw will cross the optical axis at a distance f after the Lens. The intersection is called FOCAL POINT. Here we make use of the PARAXIAL APPROXIMATION, which states that we only consider rays close to the optical axis, so we can use

$$\sin(\theta) \approx \tan(\theta) \approx \theta.$$

We will see, that in the paraxial approximation, if we have a non parallel ray incoming, it will still focus a distance f after the lens. In general, parallel rays converge after the lens in the FOCAL PLANE.

We now want to trace rays through a single lens.

By similar triangles, we have

$$\frac{\Delta}{\delta} = \frac{d}{s} =: M.$$

where M is the MAGNIFICATION of the lens. By similar triangles again, we have

$$\frac{\delta}{f} = \frac{\Delta}{d-f} \Rightarrow \frac{d-f}{f} = \frac{\Delta}{\delta} = \frac{d}{s}.$$

Hence,

$$\frac{d}{f} - 1 = \frac{d}{s} \Rightarrow \frac{1}{f} = \frac{1}{s} + \frac{1}{d}.$$

This is called the LENS EQUATION.

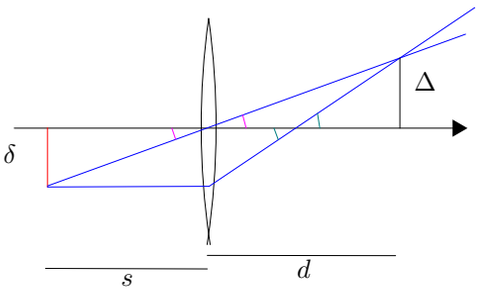


Figure 4: Deriviation of the lens equation using ray tracing.

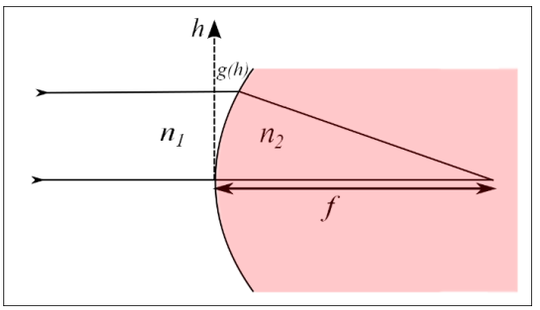


Figure 5: Setup of the surface lens.

Let us figure out how to design a simple "surface" lens. For this, we describe the profile by $g(h)$.

The optical path is given by

$$D = n_1 g(h) + n_2 \sqrt{h^2 + (f - g(h))^2}.$$

In general this is not really solvable, so we consider the paraxial approximation. So we have $h \ll f$. Since the lens should be symmetric we can assume

$$g(h) = \sum_n a_{2n} h^{2n}.$$

The simplest solution would be $g(h) = a_2 h^2$. Putting this into the expression, we get

$$\begin{aligned} D &= n_1 a_2 h^2 + n_2 \sqrt{h^2 + (f - a_2 h^2)^2} + O(h^4) \\ &= n_1 a_2 h^2 + n_2 \sqrt{h^2 + f^2 + a_2^2 h^4 - 2 f a_2 h^2} + O(h^4). \end{aligned}$$

Using that $f \gg a_2 h^2, h$ we get

$$D = n_1 a_2 h^2 + n_2 f \sqrt{\frac{h^2}{f^2} + 1 + \cancel{\frac{a_2^2 h^4}{f^2}}^{\text{neglect}} - \frac{2 a_2 h^2}{f}} + O(h^4).$$

Using the taylor expansion $\sqrt{1+x} = 1 + \frac{x}{2} + O(x^2)$, we get

$$D = n_1 a_2 h^2 + n_2 f \left(1 + \frac{h^2}{2f^2} - \frac{a_2 h^2}{f}\right) + O(h^4).$$

Reordering the terms, we get

$$D = n_2 f + \left(n_1 a_2 + \frac{n_2}{2f} - n_2 a_2\right) h^2 + O(h^4).$$

We want all paths to have the same optical path. So we want the coefficient of h^2 to vanish. This gives us

$$n_1 a_2 + \frac{n_2}{2f} - n_2 a_2 = 0 \Rightarrow a_2 = \frac{n_2}{2f(n_2 - n_1)}.$$

Unfortunately it is very annoying to make a lens with a parabolic profile. A spherical profile would be much easier.

Lenses in general are imperfect. Possible sources are:

1. Ignored higher order terms: symmetric about axis ⇒ spherical aberration

2. Off axis aberrations: coma.

3. Chromatic aberrations: The refractive index depends on the wavelength of light. So different colors focus at different points.

Lec 2

Let us look at some imaging instruments in more detail. The human eye is limited by: the size of the eye is fixed. As we have seen

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{d}.$$

For the human eye, d is fixed. Secondly, the range of focal lengths is limited. Let us consider the telescope, using lenses to help. The main problem we want to solve is that we want to see two stars subtending a small angle θ.

If we have only one lens, since the angle is small, we have

$$\theta = \frac{\delta}{s} = \frac{\Delta}{d}.$$

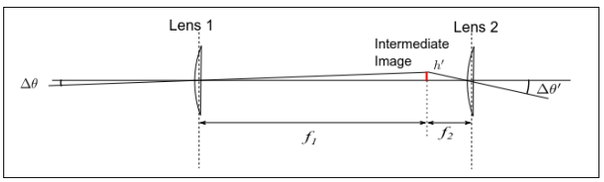


Figure 6: Telescope with two lenses.

Introducing a second lens, we first see, that image 1 is at f_1 behind lens 1, since

$$\frac{1}{f_1} = \frac{1}{s} + \frac{1}{d} \approx \frac{1}{d},$$

since s is very large. For the second lens, we then find

$$\theta_2 = \frac{\Delta}{f_2} = \frac{d\theta}{f_2} = \frac{f_1}{f_2}\theta.$$

Nowadays, we use a telescope with a mirror instead of a lens. The advantage is that we don't get chromatic aberration, since the mirror reflects all wavelengths the same way.

For looking at very small things, we typically use a microscope. Recall the magnification

$$M = \frac{\Delta}{\delta} = \frac{d}{s}.$$

So if we want to see very small things, we need to make s very small. The problem is that we need to satisfy the lens equation.

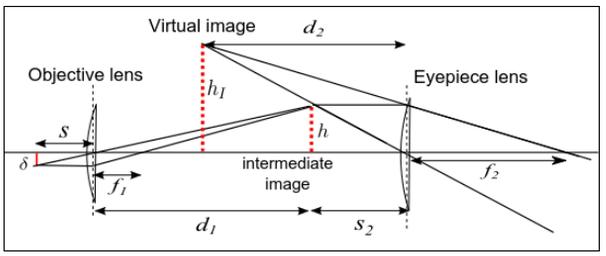


Figure 7: Microscope with two lenses.

Using a lens with $f_1 < s$ we thus get a magnification

$$h = \frac{d}{f_1}\delta.$$

In general, propagating through lenses is done, by finding the image through the first lens and then propagate that image through the second lens.

Typically in a microscope, we will use a lot of lenses. This is mainly because then we can only change the angle a little bit and hence reduce the spherical aberration. It also helps out cancelling out chromatic aberration.

1.3 Wave Optics

Starting from Maxwell's equations, we can turn them into a wave equation

$$-\nabla^2 \mathbf{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

To simplify the problem, we can decide to treat directions of $\mathbf{E}$ separately. Then for example,

$$-\frac{\partial^2 E_x}{\partial x^2} = -\mu_0 \varepsilon_0 \frac{\partial^2 E_x}{\partial t^2}.$$

This is the scalar wave equation.

Measuring the light typically involves measuring the intensity, given by

$$I = \frac{cn\varepsilon_0 |\mathbf{E}|^2}{2},$$

which is measured as power per unit area.

Let us look at the solutions of the scalar wave equation. It is given by

$$\nabla^2 V - \frac{1}{c^2}\frac{\partial^2 V}{\partial t^2} = 0.$$

We will use spherical coordinates, where the general solution is given by

$$V(r,t) = \frac{a}{r}\cos(kr - \omega t) + \frac{b}{r}\cos(kr + \omega t).$$

Here the first part represents an outgoing wave and the second part represents an incoming wave. Using complex numbers, we can write this as

$$V(r,t) = \frac{a}{r}e^{i(kr-\omega t)} + \frac{b}{r}e^{i(kr+\omega t)}.$$

We select the first term for a light source at the origin. Keep in mind that we can use complex numbers, but we can only measure real quantities. So we should always

finish with $\mathrm{Re}(V)$ or $|V|^2$. We now want to look at the propagation of light through an aperture.

For this we consider a point source at (x_s, y_s, z_s). We then place a screen at $z = 0$ with a hole in it. We then want to find the field at a point of interest (x_p, y_p, z_p).

Since we have a point source, the outgoing wave, emitted by the point source is given by

$$V_{\text{emit}} = A\frac{e^{i(ks-\omega t)}}{s},$$

where $s = \sqrt{(x-x_s)^2 + (y-y_s)^2 + (z-z_s)^2}$. To propagate this wave through the hole, we use HUYGEN'S PRINCIPLE.

Proposition 1.4: Huygen's principle

Given an incoming wave, we can write down the wavefronts. Letting each point on the wavefront be a point source of a new wave, we can then find the outgoing wave by summing up all the contributions of the new waves.

For example a planar wave front, for each point on the wave front, we can consider a point source. The outgoing wave is then given by the sum of all the contributions of the point sources. Summing up this results in again a planar wave front. As a recipe, first take the original wave

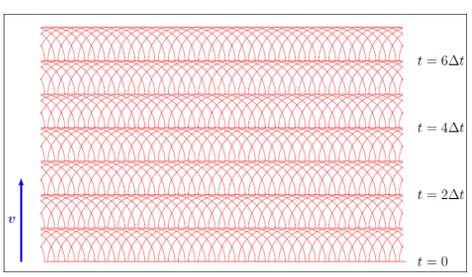


Figure 8: Huygen's principle for a planar wave front.

amplitude. The use this as a source of an outgoing wave at every position.

$$A\frac{e^{iks(x,y,z)}}{s(x,y,z)} \cdot \frac{e^{ikr}}{r},$$

where

$$r = \sqrt{(x_f - x)^2 + (y_f - y)^2 + (z_f - z)^2}.$$

Then, we sum up all the contributions of the outgoing waves.

For our problem, we constrain ourself to $x_f = x_p$, $y_f = y_p$ and $z_f = z_p$. Splitting the aperture into small pieces $\mathrm{d}x\mathrm{d}y$, we can write down the outgoing wave as

$$A\frac{e^{iks}}{s}\mathrm{d}x\mathrm{d}y \cdot \frac{e^{ikr}}{r}.$$

Summing up all the contributions, we get

$$U(P) \propto \iint \frac{e^{iks}}{s}\frac{e^{ikr}}{r}\mathrm{d}x\mathrm{d}y.$$

A more rigorous derivation would yield

$$U(P) = -\frac{iA}{4\pi}\oint \frac{e^{ik(s+r)}}{sr}\left(k\hat{r}\cdot\hat{n} - k\hat{s}\cdot\hat{n}\right)\mathrm{d}a.$$

Let us simplify the expression from Huygen's approach. First, since the aperture is small, compared to the distance to the source, we can approximate the amplitude as constant. I.e. if $s_0 \gg \lambda, \Delta_x, \Delta_y$, we can approximate $\frac{1}{s} \approx \frac{1}{s_0}$. So the incoming wave is given by

$$\frac{1}{s_0}e^{iks}.$$

Furthermore if the point of interest is far enough away, we can make the same approximation. Thus,

$$U(P) \propto \frac{A_0}{r_0 s_0}\iint e^{ik(s+r)}\mathrm{d}x\mathrm{d}y.$$

Defining the center of the aperture as $(0,0,0)$, we can write down the distances as

$$s_0 = \sqrt{x_s^2 + y_s^2 + z_s^2}, \quad r_0 = \sqrt{x_p^2 + y_p^2 + z_p^2}.$$

The interpretation of the integral is given as the amplitude is given by sum over amplitudes due to all paths. For this we simplify the expression. In general, the aperture size $\Delta_x, \Delta_y \ll |s_0|, |r_0|$. We have

$$\begin{aligned} r &= \sqrt{(x_p - x)^2 + (y_p - y)^2 + (z_p - z)^2} \\ &= \sqrt{x^2 + y^2 + z^2 + x_p^2 + y_p^2 + z_p^2 - 2(xx_p + yy_p + zz_p)} \\ &= \sqrt{x^2 + y^2 + r_0^2 - 2(xx_p + yy_p)} \\ &= r_0\sqrt{1 + \frac{x^2+y^2}{r_0^2} - 2\frac{xx_p + yy_p}{r_0^2}}. \end{aligned}$$

Using $x, y \ll r_0$, we get $\frac{x}{r_0} \ll 1 \Rightarrow \frac{x^2}{r_0^2} \lll 1$. Thus

$$r \approx r_0\left(1 - \frac{xx_p + yy_p}{r_0^2}\right).$$

Analogous calculation can be done for s. Thus we get

$$s \approx s_0\left(1 - \frac{xx_s + yy_s}{s_0^2}\right).$$

Hence,

$$U(P) = K\int_A e^{ik(s+r)}\mathrm{d}a = Ke^{ik(s_0+r_0)}\int_A e^{-ik(xX+yY)}\mathrm{d}a,$$

where

$$X = \frac{x_s}{s_0} + \frac{x_p}{r_0}, \quad Y = \frac{y_s}{s_0} + \frac{y_p}{r_0}.$$

The approximation, that the phase is linear in x and y is called the FRAUNHOFER APPROXIMATION. Lec 3

In other words, we can approximate the wavefront as planar at the aperture.

Fortunately, now the argument of the exponential is a linear function of x and y.

Integrating from $-\frac{\Delta_x}{2}$ to $\frac{\Delta_x}{2}$ and $-\frac{\Delta_y}{2}$ to $\frac{\Delta_y}{2}$, we have

$$U(P) = Ke^{ik(s_0+r_0)}\frac{2\sin(kX\Delta_x/2)}{kX}\frac{2\sin(kY\Delta_y/2)}{kY}.$$

Now the variable we're interested in is all encoded in X and Y. Using $\text{sinc}(x) = \frac{\sin(x)}{x}$, with $\lim_{x\to 0} \text{sinc}(x) = 1$, we have that the peak of the diffraction pattern occurs when $X = 0$ and $Y = 0$. This is the condition for constructive interference.

Geometrically, we can write

$$X = -\sin(\theta_0) + \sin(\theta_p) = 0 \Rightarrow \theta_p = \theta_0.$$

So the peak of the diffraction pattern occurs in the direction of the incident wave. This is a reassuring for us, since this means that light mainly travels in a straight line.

Now, the intensity is proportional to $|U|^2$, so we have

$$I(X) \propto 4\frac{\sin^2(kX\Delta_x/2)}{(kX)^2}\frac{\sin^2(kY\Delta_y/2)}{(kY)^2}.$$

Expanding the fraction and using the definition of sinc, we have

$$I(X) \propto \Delta_x^2\Delta_y^2 \cdot \text{sinc}^2(kX\Delta_x/2) \cdot \text{sinc}^2(kY\Delta_y/2).$$

So the intensity is proportional to the square of the area of the slit.

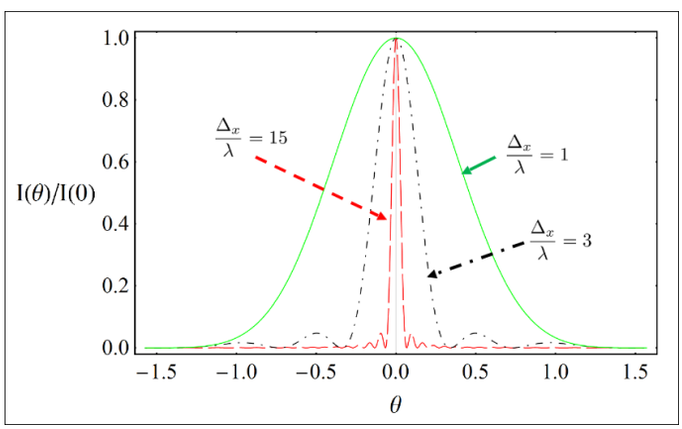


Figure 9: The diffraction pattern of a rectangular slit.

Furthermore, we notice that the sine part oscillates with X. The first width of the oscillation can be well characterized by the first zero of the sinc function, which occurs, when $\sin(k\Delta_x X/2) = 0$, or $k\Delta_x X/2 = m \cdot \pi$. Recall, that

$$X = -\sin(\theta_0) + \sin(\theta_p).$$

Simplifying with $\theta_0 = 0$, we have

$$\frac{k\Delta_x}{2}\sin(\theta_p) = m \cdot \pi \Rightarrow \sin(\theta_p) = m\frac{\lambda}{\Delta_x}.$$

The first zero occurs when $m = 1$, so we have

$$\sin(\theta_p) = \frac{\lambda}{\Delta_x}.$$

Suppose now, that we have some filter on the aperture, which is a function $F(x, y) = A(x, y)e^{i\phi(x,y)}$, where A is the amplitude and ϕ is the phase shift.

Doing the same game as before, we have

$$U = \sum A_{\text{source}} e^{iks} F(x, y) \cdot e^{ikr}.$$

In the Frauenhofer approximation, we have

$$U(P) \propto \iint F(x, y)e^{ik(Xx+Yy)}\,dx\,dy.$$

Something something Fourier transform...

In an imaging system with a lens before the aperture, we can create planar waves at the aperture. In this case, the Frauenhofer approximation is valid, and we can use the Fourier transform to compute the diffraction pattern. With a second lens, we made an imaging system. With this we can notice that every point turns into some blur.

For a telescope, we wanted to resolve the image of two stars. We now want to know how far do the stars need to be apart in order to be able to be distinguished. This is given by the Rayleigh criterion: If the centre of the image 2 falls on the first zero of image 1, then we can resolve the two stars. This means, that for the angle between the two stars, we have

$$\sin(\theta_s) > \frac{\lambda}{\Delta_x}.$$

The other problem we looked at was the microscope. In this case we have parallel light coming onto an object. We can model this with a transmission function $f(x) = \frac{1}{2}(1 + \cos(k_A x))$. Doing the Frauenhofer approximation, we have

$$\int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} \frac{1}{2}(1 + \cos(k_A x))e^{-ikXx}\,dx$$

$$= \frac{1}{2}\left[\int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} e^{-ikXx}\,dx + \int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} e^{-ikXx+ik_Ax}\,dx + \int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} e^{-ikXx-ik_Ax}\,dx\right.$$

$$\propto \frac{\sin(ka\sin(\theta))}{ka\sin(\theta)} + \frac{\sin(ka\sin(\theta) - k_Aa)}{ka\sin(\theta) - k_Aa} + \frac{\sin(ka\sin(\theta) + k_Aa)}{ka\sin(\theta) + k_Aa}.$$

With this, we find that $\frac{2\pi}{k_a} = \frac{\lambda}{\sin(\theta)}$, where $\sin(\theta)$ is the angle of the light coming from the object. This means, that we can only resolve features of size $> \lambda$, since $\sin(\theta) \leq 1$.

Lec 4

The reason we wanted that $\frac{2\pi}{k_a} = \frac{\lambda}{\sin(\theta)}$ was that we need to catch also the first side maximum of the diffraction pattern. Without this, we would loose the spatial information of the object. This is known as the Abbé limit, which states that the resolution of an optical system is limited by the wavelength of light.

1.4 Polarisation of Light

Recall the Wave equation

$$-\nabla^2\mathbf{E} + \frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2} = 0.$$

The plane wave solutions are given by

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0\cos(\mathbf{k}\cdot\mathbf{r} - \omega t).$$

Since $\nabla \cdot \mathbf{E} = 0$, we find, that $\mathbf{k} \cdot \mathbf{E}_0 = 0$, which means that the electric field is perpendicular to the direction of propagation.

For the magnetic part of the solutions, we get from $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, that

$$\mathbf{k} \times \mathbf{E}_0 = \omega \mathbf{B}_0.$$

Hence,

$$\mathbf{B}(\mathbf{r}, t) = \mathbf{B}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t).$$

With this, we note, that we can describe polarised light through the direction of the *electric* field. By linearity of the wave equation, we can also have superpositions of plane waves. For example:

$$\mathbf{E}(\mathbf{r}, t) = E_x \hat{x} \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) + E_y \hat{y} \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \alpha).$$

If $\alpha = 0$, we notice that the polarisation moves along a line, hence this is called LINEAR POLARISATION. If $E_x = E_y$ and $\alpha = \pm\frac{\pi}{2}$, we notice that the polarisation moves along a circle, hence this is called CIRCULAR POLARISATION.

If $\alpha = -\frac{\pi}{2}$ we have lefthand circular polarisation, and if $\alpha = \frac{\pi}{2}$ we have righthand circular polarisation.

Since many light sources have atoms with randomized directions, the light is often unpolarised.

When we have a surface, we will often times have a well defined direction. Given a surface, we can define the normal vector. If the normal vector lies in the same plane as spanned by $\mathbf{k}$ and $\mathbf{E}$, we call this P-POLARISATION for PARALLELE POLARISATION.

The second case is when the normal vector is in the plane spanned by $\mathbf{k}$ and $\mathbf{B}$. This is called S-POLARISATION for SENKRECHTE POLARISATION. Usually p-polarisation is depicted with a red arrow, and s-polarisation is depicted with a red dot.

Let us quickly recall linear dielectric media. If there is no electric field, then $\mathbf{D} = 0$. If we apply an electric field, the material will polarise, i.e. produce a bound charge. The relationship between the electric field and the displacement field is given by

$$\mathbf{D} - \mathbf{P} = \epsilon_0 \mathbf{E}.$$

Using, that in linear media, $\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$, we find

$$\mathbf{D} = \epsilon_0(1 + \chi_e)\mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E}.$$

Here χ_e is the electric susceptibility, and ϵ_r is the relative permittivity. The magnetic field is similar, and we have

$$\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H} = \mu_0 \mu_r \mathbf{H}.$$

In linear materials, Maxwells equations are given by

$$\nabla \cdot \mathbf{D} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}.$$

With this, we find that the wave equation in linear media is given by

$$-\nabla^2 \mathbf{E} + \mu_0 \mu_r \epsilon_0 \epsilon_r \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0.$$

Hence, the speed of light in a linear medium is given by

$$v = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{c}{n}.$$

Now, also

$$\mathbf{B}_0 = \frac{n \mathbf{k}_0 \times \mathbf{E}_0}{\omega} \quad |\mathbf{B}_0| = \frac{n |\mathbf{E}_0|}{c}.$$

When a electric wave hits a surface, we have to consider the boundary conditions.

Theorem 1.5: Boundary Conditions for Electromagnetic Waves

The boundary conditions for electromagnetic waves are given by

$$B_1^\perp = B_2^\perp, \qquad H_1^\parallel = H_2^\parallel,$$
$$D_1^\perp = D_2^\perp, \qquad E_1^\parallel = E_2^\parallel.$$

Proof. To apply the boundary conditions, we take mathematical relations and apply them to the surface. For example, by the divergence theorem, we have

$$\iiint_{\mathcal{V}} \nabla \cdot \mathbf{B} \mathrm{d}\tau = \oiint_{\partial \mathcal{V}} \mathbf{B} \cdot \mathrm{da}.$$

From Maxwells equations, we know that $\nabla \cdot \mathbf{B} = 0$. So

$$\oiint_{\partial \mathcal{V}} \mathbf{B} \cdot \mathrm{da} = 0.$$

Now, consider a pillbox with the top and bottom surface parallel to the surface, of thickness δ, where half the surface is in the medium, and the other half is on the outside.

Letting $\delta \to 0$, we find that

$$B_1^\perp = B_2^\perp.$$

The other boundary condition is derived from Stokes Theorem, which states that

$$\iint_{\mathcal{S}} \nabla \times \mathbf{H} \mathrm{da} = \oint_{\partial \mathcal{S}} \mathbf{H} \cdot \mathrm{dl}.$$

From Maxwells equations, we know that $\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$, so

$$\iint_{\mathcal{S}} \frac{\partial \mathbf{D}}{\partial t} \mathrm{da} = \oint_{\partial \mathcal{S}} \mathbf{H} \cdot \mathrm{dl}.$$

Now given a boundary, we can take a rectangular loop with two sides parallel to the surface, and two sides perpendicular to the surface. Letting the length of the perpendicular sides go to zero, notice, that the integral on the left hand side goes to zero, since the area goes to zero. From the right hand side, we find that

$$H_1^\parallel = H_2^\parallel.$$

An analogous argument can be made for the electric field, and we find that

$$D_1^\perp = D_2^\perp, \quad E_1^\parallel = E_2^\parallel.$$

□

In the following, we will focus on di-electrics with no magnetic component. So $\varepsilon_1 \neq \varepsilon_2$, but $\mu_1 = \mu_2$.

Suppose we have a p-polarised wave incident on a plane. We call the incident wave $\mathbf{E}_i$, the reflected wave $\mathbf{E}_r$, and the transmitted wave $\mathbf{E}_t$. The incident angle to the normal is θ_i, the reflected angle is θ_r, and the transmitted angle is θ_t. In the following we will assume Snell's law, however it could be derived from the boundary conditions of the magnetic field. We know that

$$\theta_r = \theta_i, \quad n_1 \sin(\theta_i) = n_2 \sin(\theta_t),$$

where $n_1 = \sqrt{\mu_1 \varepsilon_1}$ and $n_2 = \sqrt{\mu_2 \varepsilon_2}$ are the refractive indices of the two media. Since the materials are non-magnetic, we have $\mu_1 = \mu_2 = 1$, so $n_1 = \sqrt{\varepsilon_1}$ and $n_2 = \sqrt{\varepsilon_2}$.

From $H_1^\parallel = H_2^\parallel$, we find that using $\mu_1 = \mu_2 = 1$,

$$B_i + B_r = B_t.$$

Since all point in the same direction, and using $|\mathbf{B}_0| = \frac{n|\mathbf{E}_0|}{c}$, we find that

$$n_1(E_i + E_r) = n_2 E_t.$$

From $E_1^{\parallel} = E_2^{\parallel}$, we find that

$$E_i \cos(\theta_i) - E_r \cos(\theta_r) = E_t \cos(\theta_t).$$

Using $\theta_r = \theta_i$, we find that

$$(E_i - E_r)\cos(\theta_i) = E_t \cos(\theta_t).$$

Rearranging, we find that

$$\begin{aligned}
\frac{n_1}{n_2}(E_i + E_r) &= \frac{E_i - E_r}{\cos(\theta_i)} \cos(\theta_t) \\
\frac{n_1}{n_2}\left(1 + \frac{E_r}{E_i}\right) &= (1 - \frac{E_r}{E_i})\frac{\cos(\theta_t)}{\cos(\theta_i)} \\
\frac{E_r}{E_i}\left(\frac{n_1}{n_2} + \frac{\cos(\theta_t)}{\cos(\theta_i)}\right) &= \frac{\cos(\theta_t)}{\cos(\theta_i)} - \frac{n_1}{n_2} \\
&.
\end{aligned}$$

Hence,

$$\frac{E_r}{E_i} = \frac{n_2 \cos(\theta_i) - n_1 \cos(\theta_t)}{n_2 \cos(\theta_i) + n_1 \cos(\theta_t)}.$$

Doing some trigonometry, we can find that

$$\frac{E_r}{E_i} = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}.$$

This is zero, when $\theta_i + \theta_t = \frac{\pi}{2}$. Then no light gets reflected. Using Snell's law, we find that this happens when

$$\theta_i = \arctan\left(\frac{n_2}{n_1}\right).$$

This is known as the Brewster angle.

This effect is for example useful when taking holidays at the sea. The light reflected from the water is mainly s-polarised, so if you wear polarised sunglasses, you can reduce the glare from the water.

Lec 5

1.5 Spectroscopy

We now want to discuss how different colors of light can be used and how we can separate colors. One device to do that is the prism. This is the idea behind spectroscopy. In general, spectroscopy relies on separating colours/wavelength/frequency into different spatial features.

To achieve this we want to use interference. For this we split up the light and send it down different paths. In the prism, the light is split up into different colors because of the different refractive indices for different wavelengths. By Snell's law, we know, that the angle of refraction then changes with the wavelength.

The first interference spectrometer we look at is a Grating Spectrometer. For this, we have a source which sends out some noisy light. We use a lens to focus the light into a slit. This is used to limit, where light comes from. In addition we have a focal lens a distance f away from the slit. This will produce planar waves after slit. In the path of the light, we put a grating. It is tilted at an angle θ_i to the optical axis. Of the grating, there will be some light reflected at an angle θ_i. Afterwards, we need another lens to focus the light onto a given point on the screen. The question is now, at which angles we see interference and at which angles we see destructive interference.

We can also implement a reference, with a known source. Then we can compare the interference pattern of the known source with the unknown source.

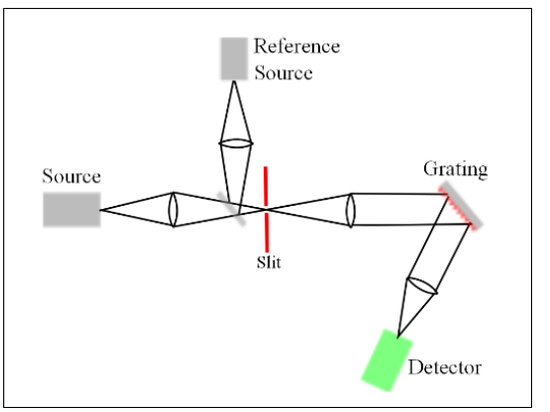


Figure 10: Grating Spectrometer

The grating is a surface with places that can reflect light and places that cannot. In general, we want to have a periodic structure. Since light is a plane wave, we can assume, that all light is coming from the same angle. By the wavelike nature of light, we can have a outgoing angle of θ_j which is not nesessarily equal to the incoming angle θ_i.

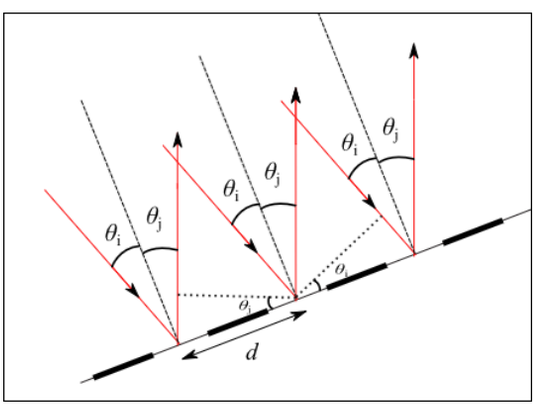


Figure 11: Grating

We now want to find the phase difference of the outgoing light. The phase shift is given by $\delta\phi = k\Delta x$, where k is the wave number and Δx is the path difference.

The extra distance the has to travel between slit n and $n+1$ is given by

$$\Delta x_{\text{incoming}} = d\sin(\theta_i) \quad \text{and} \quad \Delta x_{\text{outgoing}} = d\sin(\theta_j).$$

So in total, $\Delta x = d(\sin(\theta_j) - \sin(\theta_i))$. The phase difference is then given by

$$\delta\phi = kd(\sin(\theta_j) - \sin(\theta_i)).$$

To see light under the angle θ_j, we need to have constructive interference. This happens, that $\delta\phi = 2\pi\mathbb{Z}$.

Over the full grating, we have an initial amplitude $A(0)$, and we sum up over all the slits.

$$\begin{aligned}A(\theta_j) &= A(0) + A(0)e^{i\delta\phi} + A(0)e^{2i\delta\phi} + \cdots + A(0)e^{i(N-1)\delta\phi} \\ &= A(0)\sum_{n=0}^{N-1} e^{in\delta\phi} \\ &= A(0)\frac{1-e^{iN\delta\phi}}{1-e^{i\delta\phi}} \\ &= A(0)\frac{e^{iN\delta\phi/2} - e^{-iN\delta\phi/2}}{e^{i\delta\phi/2} - e^{-i\delta\phi/2}} e^{i(N-1)\delta\phi/2} \\ &= A(0)\frac{\sin(N\delta\phi/2)}{\sin(\delta\phi/2)} e^{i(N-1)\delta\phi/2}.\end{aligned}$$

Thus, for the intensity we have

$$I(\theta_j) = |A(\theta_j)|^2 = |A(0)|^2 \frac{\sin^2(N\delta\phi/2)}{\sin^2(\delta\phi/2)}.$$

When the denominator goes to zero, we get peaks of height N^2. All other features have height ≈ 1. Notice, that since the period of constructive interference is dependent on the wavelength, we can use this to separate different wavelengths. We now want to find a condition like the Rayleigh criterion, but for spectral resolution. I.e. when does the peak of one color lie on the first zero of the other color.

For this, we can write

$$\delta\phi = 2\pi\frac{d}{\lambda}(\sin(\theta_j) - \sin(\theta_i)).$$

For peaks, we can have seen that $\delta\phi = 2\pi m$, where m is an integer. Note, that at these points, the numerator is also zero. So the limit of the function is finite. Note, that the numerator is zero if $k\pi = N\delta\phi/2$, where k is an integer. Since we have a peak if $k = Nm$, the nearest zero is at $k = Nm + 1$. Thus, the nearest zero is at

$$\frac{N\delta\phi}{2} = (Nm+1)\pi.$$

The condition we want is that when we have λ_1, f_1 constructively interfering at an angle θ_j, we need to get destructive interference for λ_2, f_2 at the same angle. This means, that we need

$$\begin{aligned}\delta\phi_1 &= k_1 d(\sin(\theta_j) - \sin(\theta_i)) = 2\pi m \\ \delta\phi_2 &= k_2 d(\sin(\theta_j) - \sin(\theta_i)) = 2\pi m + \frac{2\pi}{N}.\end{aligned}$$

Combining and rearranging, we get

$$k_2 - k_1 = \frac{2\pi}{Nd(\sin(\theta_j) - \sin(\theta_i))}.$$

In terms of frequencies, we have $k = 2\pi f/c$. Thus, we can write

$$\delta f = \frac{c}{Nd(\sin(\theta_j) - \sin(\theta_i))}.$$

Notice, that Nd corresponds to the total length of the grating w. Since sine is bounded by 1, we find the maximal value of $Nd(\sin(\theta_j) - \sin(\theta_i))$ is $2w$. Thus, the smallest frequency difference we can resolve is

$$\delta f_{\min} = \frac{c}{2w}.$$

This corresponds to the time it takes light to travel the longest path difference we can introduce.

To make a long path difference, we can for example use a Michelson Spectrometer. Again, we have a source, a slit and a lens. The light is the split by a beam splitter. At the ends of both paths, there are mirrors. The light is reflected back and recombined at the beam splitter. The light is then focused onto a screen.

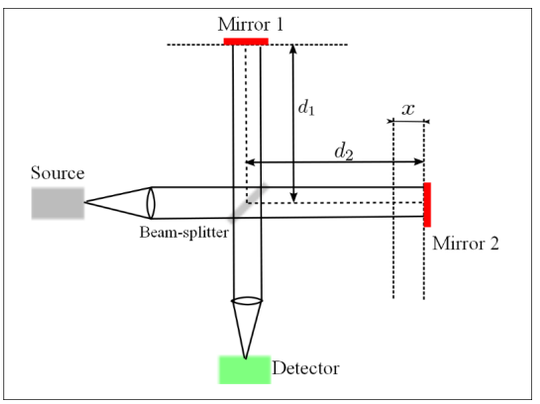


Figure 12: Michelson Spectrometer

The difference in phase for light travelling via mirror 2 versus mirror 1 is given by

$$\delta\phi = k\Delta x = 2k(d_2 - d_1).$$

Using a complex wave at the detection, we can find a wave of the form

$$A = \frac{A_0}{2}\left(1 + e^{i\delta\phi}\right).$$

Defining $x \coloneqq d_2 - d_1$, we can write

$$I(x) = \frac{|A|^2}{|A_0|^2} = \frac{1}{2}|1 + \cos(\delta\phi)| = \frac{1}{2}(1 + \cos(4\pi x\frac{f}{c})).$$

We can rewrite the amplitude of the original light as seen as

$$I(x) = \frac{A_0^2}{2}(1 + \cos(\delta x)) = A_0^2\cos^2(\delta x/2).$$

This is one, if $\delta x = \pi n$ and zero if $\delta x = \pi(n + 1/2)$. The distance between two zeros is then $\Delta x = \pi$.

So for dark spots, we require

$$2x = (n + \frac{1}{2})\lambda.$$

For bright spots, we require

$$2x = n\lambda.$$

Performing the experiment, we expect to see either a bright or a dark but fringe. However, we can see stripes. This implies that one mirror is tilted. If the mirror was perfectly aligned, we should see rings, if the beam is not perfectly collimated.

We think about power density in a frequency range from f to $f+\delta f$. The power density between f and $f+\delta f$ is given by $S(f)\delta f$. We claim that the intensity in a Michelson

spectrometer is given by as a sum of contributions from all frequencies. Thus,

$$I_{\text{tot}}(x) = \text{const} \cdot \sum \text{all} I(x) \text{from diffrent} f.$$

Letting $\delta f \to 0$, we can write this as an integral

$$I_{\text{tot}}(x) = \text{const} \cdot \int S(f) I(x) df \propto \int S(f) \frac{1}{2}(1+\cos(4\pi x \frac{f}{c})) df.$$

Expaning, we get

$$I_{\text{tot}}(x) = I_0 + \frac{1}{2}\int_0^\infty S(f) \mathrm{d}f \cos(4\pi x \frac{f}{c}).$$

So $(2I_{\text{tot}}(x) - I_0)$ is the Fourier transform of $S(f)$. Inverting the fourier transform, we get

$$S(f) = \int_0^{x_{\max}} (2I_{\text{tot}}(x) - I_0) \cos(4\pi x \frac{f}{c}) dx.$$

Lec 6

Note, that in the sum the different frequencies will have cross terms, however since we measure over a long time scale, these will average to zero. Furthermore, we can only integrate to $x_{\max}$, since we cannot measure from 0 to ∞. Thus, we can write

$$\begin{aligned} S_{\text{meas}}(f_1) &= \int_0^{x_{\max}} dx\, (2I_0 - I_0) \cos(4\pi x \frac{f_1}{c}) \\ &\propto \int_0^\infty \mathrm{d}f\, S(f) \int_0^{x_{\max}} dx\, \cos(4\pi x \frac{f}{c}) \cos(4\pi x \frac{f_1}{c}). \end{aligned}$$

Using the identity $\cos(a)\cos(b) = \frac{1}{2}(\cos(a-b)+\cos(a+b))$, we can write

$$\begin{aligned} S_{\text{meas}}(f_1) &\propto \int_0^\infty \mathrm{d}f\, S(f) \int_0^{x_{\max}} dx\, \frac{1}{2}\left(\cos(4\pi x \frac{f-f_1}{c}) + \cos(4\pi x \frac{f+f_1}{c})\right) \\ &\propto \int_0^\infty \mathrm{d}f\, S(f) \left(\frac{\sin(2\pi x_{\max}\frac{f-f_1}{c})}{2\pi x_{\max}\frac{f-f_1}{c}} + \frac{\sin(2\pi x_{\max}\frac{f+f_1}{c})}{2\pi x_{\max}\frac{f+f_1}{c}}\right) \\ &= \int_0^\infty \mathrm{d}f\, S(f) \left(\text{sinc}(2\pi x_{\max}\frac{f-f_1}{c}) + \text{sinc}(2\pi x_{\max}\frac{f+f_1}{c})\right) \end{aligned}$$

Assuming that we started with a single frequency, $S(f) = I_0\delta(f-f_0)$, where f_0 is the source frequency, and f_1 is the frequency we are measuring. For $f_0 \approx f_1$, we can ignore the second term, and we get

$$S_{\text{meas}}(f_1) \propto I_0 \left(\text{sinc}(2\pi x_{\max}\frac{f_0 - f_1}{c})\right).$$

Because of the Rayleigh criterion, we will say that we describe the broadness of the peak by the first zero of the sinc function. This is located at

$$\frac{4\pi}{c} x_{\max} \underbrace{(f_0 - f_1)}_{\delta f} = \pi \Rightarrow \delta f = \frac{c}{4x_{\max}}.$$

Note, that $4x_{\max}$ is two times the maximum path difference, where the factor two appears because the light travels to the mirror and back. Thus, we can write

$$\frac{\delta f}{f} = \frac{\lambda}{4x_{\max}}.$$

In conclusion, the resolution is set by the longest path difference one can create.

Suppose now, we have a source with two different frequencies f_1 and $f_2 = f_1 + \delta f$. The total intensity is now

$$I_{\text{tot}}(x) = \frac{1}{2}(1 + \cos(2\pi x \delta f/c)\cos(2\pi x(2f_1 + \delta f)/c)).$$

The second term produces far fringes. The first term moves rather small compared, so we call it the ENVELOPE. The fast fringes looks like a normal single frequency at frequency $2f_1 + \delta f$. The envelope has a period of $c/\delta f$. In order to see we have two different frequencies, we need to see the change in contrast from its maximum to its minimum. For this, we need at least half a period of the envelope. Thus, we need to have a path difference of at least

$$D = \frac{2\pi x}{c}\delta f = \frac{\pi}{2} \Rightarrow \delta f = \frac{c}{4\delta x}.$$

Since $D = 2x$, we again arrive at

$$\delta f = \frac{\lambda}{2D}.$$

To get a long path difference, on limited space, we can use a FABRY-PEROT ETALON. A etalon is essentiall 2 flat parallel reflective surfaces at a distance d. Strictly speaking, the etalon has a fixed distance.

Let's determine what happens to the phase of light when it is reflected back and forth between the two surfaces. The incoming wave has the form $E_{\text{in}} = A_0 e^{i(kx-\omega t)}$. We will assume that the mirrors are the same. Then, light can be transmitted $T = t^2$ $E_{\text{in}} \to tE_{\text{in}}$. If we instead reflex, $R = |e^{i\phi}r|^2$, $E_{\text{in}} \to re^{i\phi}E_{\text{in}}$. Lastly light can travel a distance d, $E_i n \to e^{ikd}E_{\text{in}}$. Let us just consider the light that is transmitted through the etalon. The first transmitted light is

$$A_0' = A_0 t e^{ikd} t = A_0 t^2 e^{ikd}.$$

What if we reflect twice, then we have

$$A_1 = A_0 t e^{ikd} r e^{i\phi} e^{ikd} r e^{i\phi} t = A_0 t^2 r^2 e^{ikd} e^{i2(kd+\phi)}.$$

So we get

$$\frac{A_1'}{A_0'} = R e^{i2(kd+\phi)}.$$

So

$$An' = A_0 t^2 e^{ikd} (Re^{i\delta\phi})^n.$$

So for the output power, we get

$$A_{\text{out}} = A_0 t^2 e^{ikd} \sum_{n=0}^{\infty} (Re^{i\delta\phi})^n = A_0 t^2 e^{ikd} \frac{1}{1 - Re^{i\delta\phi}}.$$

So we also find

$$\begin{aligned} \frac{I}{I_0} &= \frac{T^2}{1 + R^2 - 2R\cos(\delta\phi)} \\ &= \frac{(1-R)^2}{(1-R)^2 + 2R(1-\cos(\delta\phi))} \\ &= \frac{1}{1 + \frac{2R}{(1-R)^2}(1-\cos(\delta\phi))} \\ &= \frac{1}{1 + \frac{4R}{(1-R)^2}\sin^2(\delta\phi/2)}. \end{aligned}$$

This tells us, that the maxima of this function are, when $\delta\phi = 2\pi\mathbb{Z}$. For $R \to 1$, we see that the peaks get sharper and sharper. The peak spacing is given by $\delta\phi = 2\pi$.

We can now combine a Fabry-Perot etalon with a diffraction grating. The etalon will produce sharp peaks, and the diffraction grating will separate the different wavelengths. Lets focus on the etalon first. After the etalon, notice, that the peak condition is

$$2\pi n = 2kd + 2\phi.$$

Since 2ϕ is a constant, we see that we get a train of peaks, one per order if

$$\Delta d = \frac{\pi}{k} = \frac{\lambda}{2}.$$

This also explains why we need a grating. It serves as a large scale filter, so that we can separate the different orders.

So if we have two frequencies. For different k, we will need different d.

So once again, given two frequencies, how far can we measure? The limit is given, if one order falls exactly where the second order is of the other frequency. This is given by

$$2(k_2 - k_1)d = 2\pi \Rightarrow \Delta f = \frac{c}{2d}.$$

This quantity is called the Free spectral range. It describes the maximum frequency difference, that can be measured with a Fabry-Perot etalon. Going back to the airy function, we define the Full width half maximum (FWHM) as the width of the peak at half its maximum. The free spectral range tells us how far apart the resonant frequencies of a cavity are.

Lec 7

We now want to determine how many frequencies we can tell apart. For this, let us look at the airy function again. For half the intensity, we require

$$1 = \frac{4R}{(1-R)^2}\sin^2\left(\frac{\delta}{2}\right).$$

Doing a small angle approximation, we have

$$1 = \frac{4R}{(1-R)^2}\left(\frac{2\pi\delta f}{c}d\right)^2.$$

So

$$\delta f = \frac{c}{2d}\frac{1-R}{2\pi\sqrt{R}}.$$

Since this is only one side of the peak, we have to multiply by 2 to get the FWHM:

$$\delta f_{FWHM} = \frac{c}{2d}\frac{1-R}{\pi\sqrt{R}} = FSR\underbrace{\frac{1-R}{\pi\sqrt{R}}}_{\frac{1}{F}}.$$

Where we define the Finesse as

$$F = \frac{FSR}{\delta f_{FWHM}} = \frac{\pi\sqrt{R}}{1-R}.$$

Note, that the Finesse only depends on the reflectivity of the mirrors, and not on the length of the cavity. The Finesse tells us how many peaks we can resolve in a given frequency range.

The Finesse tells us how many peaks we can resolve in a given frequency range. It is a measure of the longest path in the etalon, which can be estimated by $d \cdot F$.

Experimentally, we can see that a Fabri-Perot cavity has rings as an interference pattern. This is why we need a slit to select a single mode. The intuition is that the round trip path length depends on the incoming beam angle. This is because one round trip actually takes a distance $\frac{2d}{\cos\theta}$, where θ is the angle of incidence. Furthermore, there is an extra path length of

$$d'' = 2d\tan\theta\sin\theta.$$

The rings are bright, if $2d\cos(\theta) = m\lambda$. With increasing θ, the rings get closer together.

2 Photons

2.1 Photoelectric effect

Consider the following setup. A mercury lamp passes through a filter and hits a zinc plate, which is connected to an electrometer, measuring static charge. If there is no charge, we excpect no change. If we have negative charges on the electrometer, we see, that the plate gets discharged when light shines on it. However, using a glass plate as a filter, no discharge occurs. If the plate was instead of quartz, the discharge occurs. The difference between them is, that glass has a transmission curve between 300 and 2000 nm, while quartz has a transmission curve between 140 and 4000 nm. So the effect has to do with UV or IR light.

Using $I \propto |\mathbf{E}\cos(kz-\omega t)|^2$, we would excpect the electrons in the plate to move with

$$x = \frac{eE_0}{m\omega^2}\cos(kz-\omega t) \Rightarrow v = -\frac{eE_0}{m\omega}\sin(kz-\omega t).$$

Hence,

$$\langle E_{kin}\rangle = \frac{1}{2}m\langle v^2\rangle = \frac{e^2E_0^2}{4m\omega^2}.$$

This grows with intesity, but also, higher frequencies imply lower kinetic energy. But this is *not* what we observe. But clearly, all of this is wrong, because the mercury lamp has an emission along 254nm to 600nm. So something is wrong with our classical picture.

This is described in the PLANCK EXPERIMENT. For this we have a mercury lamp, a selective frequency filter and a detector. The detector here is a PHOTOTUBE. For this, we have a amperemeter connected to a cathode and an anode. We also put this setup in a vacuum. We also have a variable voltage source, which can be used to apply a voltage between the cathode and the anode. Let $U = V_A - V_C$. If this is less than 0, it slows down the electrons, if its bigger, it accelerates them. If this works, the electrons leave the cathode with E_{kin}. It then arrives at the anode if $E'_{\text{kin}} \geq 0$ where E'_{kin} is the energy at the anode. From energy conservation, we have

$$E_{\text{kin}} - eV_c \overset{!}{=} E'_{\text{kin}} \geq 0 \Rightarrow E_{\text{kin}} \geq e|V_c|.$$

Hence there should be some breaking voltage U_B where $e|U_B| > E_{\text{kin}}$.

Measuring the breaking voltage for different wavelengths, we find that the braking voltage is linear. In the following we will use ν for the frequency of the light. The slope of the linear fit is given by

$$\frac{\Delta U_B}{\Delta\nu} \approx 4\times10^{-15}\,\text{V s}..$$

Hence,

$$\frac{\Delta E_{\text{kin}}}{\Delta\nu} = 6.57\times10^{-34}\,\text{J s} \approx h.$$

Where h is the PLANCK CONSTANT.

If we instead measure the intensity of the current, we find the shape of an S-curve.

Max Planck in 1900 sugessted, that electromagnetic waves can only have "packets" of energy. 5 years later, Einstein suggested, that in each packet, called QUANTA, there is an energy $E = h\nu$. This quanta later was called a PHOTON. Now how does this explain our observations? For the breaking voltage depending linear on frequency, not intensity, we can see that if one photon is absorbed by one electron, then $h\nu = E_{\text{kin}} + const.$ Denoting the constant as $h\nu_{\text{min}}$, we have

$$E^B_{\text{kin}} = h\nu - h\nu_{\text{min}}.$$

If we look at the limiting case, we find

$$E^B_{\text{kin}} = h\nu_{\text{min}} = eV_W.$$

Here W is called the WORK FUNCTION. It describes the minimal energy needed to remove an electron from a solid. The work function depends on the material and surface properties.

For the observed S-curve, the power is the photon energy times the flux

$$P = h\nu\cdot\dot{N}p\nu \Rightarrow I_S = e\dot{N}p\nu = \frac{eP}{h\nu}.$$

This fixes the saturation and the shape of the S-curve is a material property.

As an application, Photomultiplier tubes (PMT) are used to detect single photons. For this, after the cathode, we place DYNODES. Theses are charge multipliers via acceleration. From one dynode to the next, there is a voltage difference of about 100 V. So 1 electron can knock out several electrons from the next dynode. This is useful since the current of one electron is too small to be measured. The PMT can have 10-12 dynodes, so the current can be amplified by a large factor. Such PMTs are for example used in ICEcube, which won the nobel prize 2 days ago.

So what about the inverse process? Suppose a free electron with kinetic energy E_{kin} hits a metal surface. The kinetic energy is then converted into a photon.

For visible light, like $\lambda = 500$ nm, we require $h\nu = 2.5$ eV. For x-rays, with $\lambda = 0.1$ nm, we require $h\nu = 12$ keV. This is doable, but quite hard. For something even more energetic like gamma rays with $\lambda = 1pm$, we require $h\nu = 1$ MeV. This is too hard to achieve with electrons.

This is indeed, how we can produce x-rays. For this we have electrons accelerated through a vauum tube, which then hit a metal target (anode). For the electron source we can use a heated FILAMENT. If it's hot enough for electrons to overcome the work function, they can leave the filament. This is called THERMIONIC EMISSION. After the electrons get accelerated, they hit the anode. This deceleration creates radiation. In reality, about 1% of the energy is converted into x-rays, the rest is heat. For this we need to water cool the anode. The emission is

$$h\nu_{\text{max}} = eU + W \approx eU.$$

From this, we get

$$\lambda_{\text{min}} = \frac{hc}{eU}.$$

Looking at the emission spectrum, we notice a weird feature for high energies. This is due to interference for x-rays. For visible light, we had $d\sin\theta = m\lambda$. Since we

needed $\sin\theta \leq 1$, we needed λ and d in the same size. But now, if λ is 0.1 nm, we need d to be of the same size. This is quite hard to produce. We can replace our slits by atoms, so we are talking about a crystal. In order to have coherent light, we need to have elastic scattering, i.e. the electrons should not lose energy.

Lec 8

Let us look how the X-rays can interact with the bound electrons in the atoms. For this, we can approximate the binding potential of the electrons in the atom as a harmonic oscillator potential. As such, we can characterize the electron as a mass on a spring with resonance frequency ω_0. The X-rays then drive this mass. We get the equation of motion

$$m\ddot{x} + m\omega_0^2 x = -eE\cos(\omega t).$$

Hence,

$$x_0 = \frac{eE}{m(\omega^2 - \omega_0^2)}.$$

In general, we assume $\omega > \omega_0$, so $x_0 > 0$. This corresponds to a dipole, which radiates at driving frequency ω with fixed phase. Since we did some approximations, this result is only valid if x_0 stays in the harmonic region and the incoming field should not go too close to the resonance frequency ω_0. This model is called Bragg scattering.

Recall that for a diffraction grating, we have the condition for constructive interference

$$n\lambda = d\sin(\theta).$$

For Bragg scattering, the interference condition is given by

$$2a\sin(\theta) = n\lambda.$$

Where a is the distance between the planes of atoms in the crystal and θ is the angle of incidence. To detect the signal, we place the detector at the angle 2θ. In that sense, the crystal acts like a mirror.

However, a crystal has different planes. This allows us to have even more interference conditions.

Until now, we have considered the photon as elastic. Now we will consider it as inelastic. This is called Compton scattering. For this we want to give the photon a momentum. For this, we use the energy-momentum relation for the photon

$$E^2 = p^2c^2 + m^2c^4.$$

Since the photon is massless, we have $E = pc$. Hence, the momentum of the photon is

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}.$$

Since the photon is inelastic, we want to behave the electron like a free particle. Hence, $h\nu$ has to be bigger than the binding energy of the electron. To make this work, we typically need γ-rays. The typical experiment to look at this is to have a source like ^{137}Cs, which emits γ-rays with energy $E = 662$ keV. The rays then hit a copper rod, which scatters the rays. The rays then hit on a NaI detector, after which a photomultiplier is placed. We use NaI because Iodine is heavy which stops the gamma rays and then reemitts them as visible, by the PMT detectable light.

To determine the scattering, consider a photon with energy $E = h\nu$ hitting an electron at rest. The photon is scattered at an angle θ with energy $E' = h\nu'$.

Energy conservation gives us

$$h\nu + m_e c^2 = h\nu' + \sqrt{p_e^2c^2 + m_e^2c^4}.$$

Meanwhile from momentum conservation, we have

$$\begin{aligned}\frac{h\nu}{c} &= \frac{h\nu'}{c}\cos(\theta) + p_e\cos(\phi)\\ 0 &= \frac{h\nu'}{c}\sin(\theta) - p_e\sin(\phi).\end{aligned}$$

Squaring and adding the two momentum equations, we get

$$h^2\nu^2 + h^2\nu'^2 - 2h^2\nu\nu'\cos(\theta) = p_e^2c^2.$$

Written more compactly, we have

$$p_e^2c^2 = h^2\nu^2 + h^2\nu'^2 - 2h^2\nu\nu'\cos(\theta).$$

And

$$p_e^2c^2 + m_e^2c^4 = [h(\nu - \nu') + m_ec^2]^2.$$

Together, this gives

$$\begin{aligned}-2h^2\nu\nu'\cos(\theta) &= -2h^2\nu\nu' + 2h(\nu - \nu')m_ec^2\\ h^2\nu\nu'(1 - \cos(\theta)) &= h(\nu - \nu')m_ec^2\\ \frac{h}{m_ec^2}(1 - \cos(\theta)) &= \frac{1}{\nu'} - \frac{1}{\nu} = \frac{\lambda' - \lambda}{c}.\end{aligned}$$

We usually call $\lambda_C = \frac{h}{m_ec}$ the Compton wavelength. It describes the equivalent wavelength of an electron. Hence, we have

$$\lambda' - \lambda = \lambda_C(1 - \cos(\theta)).$$

In terms of frequency, we have

$$\frac{\nu'}{\nu} = \frac{1}{1 + \frac{\nu\lambda_C}{c}(1 - \cos(\theta))}.$$

The compton wavelength is $\lambda_C = 2.43 \times 10^{-12}$ m. At 90 degrees, we see that $\delta\lambda = \lambda_C$. This also, what compton measured in his original experiment.

In conclusion, we see that $\delta\lambda$ is maximal for back scattering, i.e. $\theta = 180°$. Furthermore, $\delta\lambda$ is independent of the incoming wavelength λ. With gamma rays, at 90 degrees, we see that more than 50% of the energy is transferred.

3 Atoms and their Composition

Let us look at the periodic table. In it, the atoms are ordered by their atomic number Z. We denote atoms with the notation ${}^{A}_{Z}X$ where A is the mass number, Z is the atomic number and X is the chemical symbol. To find relative masses, historically, the ideal gas law $PV = Nk_BT$ was used, which tells that every gas contains the same number of particles.

One can then fill up two containers with different gases at the same temperature and pressure. Then, one can measure the mass of the two containers. This gives a ratio of the masses. So to get absolute masses, we only need one as a reference. The idea is to take a perfect crystal, then measure the latice spacing (a) with X-ray diffraction. Then

$$m_{\text{atom}} = \frac{a^3\rho}{n_{\text{cell}}},$$

where $\rho = m/V$ is the density. Up to 2019 this was done by comparing it to a mass standard. Since 2019, the mass is defined by the Planck constant h.

A better way to determine masses is to do mass spectrometry. This is useful, since different isotopes have different masses. The setup is as follows. We have a source, which emits atoms. We then ionize the atoms. Then the atoms fly through a capacitor inside a magnetic field. We expect, with $\mathbf{E} = E\mathbf{e}_y$ and $\mathbf{B} = B\mathbf{e}_y$ and $\mathbf{v} = v\mathbf{e}_z$ as starting velocity, that the Lorentz force is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = q(E\mathbf{e}_y - vB\mathbf{e}_x).$$

So the equations of motion in the y direction is

$$m\ddot{y} = qE => y = \frac{qE}{2m}t^2.$$

For x, assuming the length of is small, we have

$$m\ddot{x} = -qvB \Rightarrow x = -\frac{qvB}{2m}t^2.$$

The time spent in the field is $t = \frac{L}{v}$. Hence,

$$y = \frac{qE}{2m}\frac{L^2}{v^2}, \quad x = -\frac{qvB}{2m}\frac{L^2}{v^2} = -\frac{qBL^2}{2mv}.$$

Hence,

$$\frac{1}{v} = -\frac{x^2 m}{qBL^2} \Rightarrow y = \frac{2Ex^2}{B^2L^2} \cdot \frac{m}{q}.$$

Hence, we expect a parabolic curve. Also, every ion with the same $\frac{m}{q}$ ratio will land on the same parabola. The speed only sets where it lands. If its fast, its near the zero point, if its slow, it lands further away. This was used by Thomson in 1913 to discover the first isotopes of stable elements. One problem of the method is that you have a parabolic curve, which means the intensity per point is low. Ideally, we want to have all $\frac{q}{m}$ ions to land on the same point.

The idea here is to use a radial **E**-field. The ion will the follow a circle.

$$r = \frac{mv^2}{qE}.$$

The magnetic field is then dispersive in momentum with

$$r = \frac{mv}{qB}.$$

So we can do direction focusing and velocity focusing. So now we can determine the mass very well. What is now interesting is the size and charge distribution of atoms.

Lec 9

4 Statistical Mechanics

The goal of statistical mechanics is to connect microscopic descriptions to macroscopic variables. For example, we might have a lot of particles in a gas, each with its own position and velocity. We want to connect this to the temperature of the gas.

At the microscopic level, we have deterministic equations of motion. So if we know the position $\mathbf{q}_i$ and momentum $\mathbf{p}_i$ of each particle, we can use Newton's laws to predict the future values of $\mathbf{q}_i$ and $\mathbf{p}_i$ for all time. The issue is, that for a gas at room temperature and pressure, we have $N_p \sim 10^{23}$ particles. This is too much information to store and manipulate. So we have to ask ourselves how to predict bulk properties of the gas from the microscopic equations. To do this, we use statistical methods.

Let us first look at discrete probability distributions. The most famous example is probably the dice. It has 6 outcomes, each with probability 1/6. Another more complex example is a coin toss of 4 fair coins. There are $2^4 = 16$ outcomes. However we only care about the total amount of heads. So we can group the outcomes into 5 groups: 0 heads, 1 head, 2 heads, 3 heads, and 4 heads. The probability of getting 0 heads is 1/16, the probability of getting 1 head is 4/16, the probability of getting 2 heads is 6/16, the probability of getting 3 heads is 4/16, and the probability of getting 4 heads is 1/16. This is called the binomial distribution.

For continuous probability distributions, the probability to find a particle between x and $x+dx$ is given by $p(x)dx$. Note, that $p(x)$ is not a probability! It is a probability density, which becomes a probability when multiplied with an interval.

An important quantity is the EXPECTATION VALUE of a quantity. For example, a dice has the outcomes $X = 1, 2, 3, 4, 5, 6$ with probabilities $p(X) = 1/6$. The expectation value is given by

$$\langle x \rangle = \frac{1 + 2 + \cdots + 6}{6} = \sum XP(X) = 3.5.$$

In the continuous case, the expectation value is given by

$$\langle x \rangle = \frac{\int x \cdot (p(x)\mathrm{d}x)}{\int p(x)\mathrm{d}x}.$$

Consider a single particle in a box. Suppose the particle moves with a velocity v_z to wards a wall. The particle will hit the wall and bounce back. For this the wall excerts a force $F_z(\text{wall})$. The impulse of collision relates to

$$\delta p = -2mv_z.$$

We assume collisions to be repetitive at time intervals $\Delta t = \frac{2L}{v_z}$. The average force on the wall is then given by

$$\langle F_z(\text{wall}) \rangle = \frac{\delta p}{\Delta t} = -\frac{mv_z^2}{L}.$$

In a gas, we have atoms with velocity between v_z and $v_z + \mathrm{d}v_z$, assigned to the probability $p(v_z)\mathrm{d}v_z$. The average force on the wall is then given by

$$\left\langle F^{\text{tot}} \right\rangle = N \int \frac{mv_z^2}{L} p(v_z)\mathrm{d}v_z = \frac{N \cdot m}{L} \langle v_z^2 \rangle.$$

Since $F = pA$, we can write this as

$$pA \cdot L = Nm\langle v_z^2 \rangle \Rightarrow pV = Nm\langle v_z^2 \rangle.$$

We want to use statistical physics to show that

$$\frac{k_B T}{2} = \frac{m\langle v^2 \rangle}{2}.$$

Definition 4.1: Macrostate

Macrostates are defined by bulk variables such that P, V, E, N. Also probability distributions are macrostates.

Definition 4.2: Microstate

Microstates are a description the microscopic variables such that $\mathbf{q}_i, \mathbf{p}_i$ for all particles $i = 1, \ldots, N$.

Returning to our coin example with 4 coins. We have $N_{\text{coins}} = 4$. Each coin is either heads or tails. A microstate is for example: "Coin 1 is heads". A macrostate might be "1 coin is heads". Notice, that the macrostate does not tell us which coin is heads. There are many microstates that correspond to the same macrostate. For example, "Coin 1 is heads" and "Coin 2 is heads" are two different microstates that correspond to the same macrostate of "1 coin is heads".

A different example is a system of particles, which take energies $n \cdot \varepsilon$. Given 7 particles at total energy 3ε, we can have for example the two microstates:

$$\text{Microstate 1: } (0, 0, 0, 0, 0, 3\varepsilon, 0)$$
$$\text{Microstate 2: } (0, 0, \varepsilon, 0, \varepsilon, \varepsilon, 0).$$

In this course we will define these as separate macrostates as the probability distributions are different.

4.1 The Boltzmann Distribution

We want to look at equilibrium distributions. For this we go back to the coin-flipping game. The coins start in some initial state. At every step of the game, choose one coin at random and flip it. If we interrupted the game after a long time, what can someone predict about the macroscopic variable "Number of heads"?

Consider the macrostate $N_H = 4$. This gives us the probabilistic condition

$$P(3|4) = 1.$$

Since regardless of the microstate, we will always have 4 heads. If we start with $N_H = 3$, we get the probabilistic values

$$P(4|3) = \frac{1}{4}, \quad P(2|3) = \frac{3}{4}.$$

So now we can write down the change in probability of having 4 heads at time $t+1$ given the probabilities at time t:

$$\begin{aligned} P_{t+1}(N_H = 4) &= P(4|4)P_t(N_H = 4) + P(4|3)P_t(N_H = 3) \\ \Rightarrow \Delta P(4) &= P_{t+1}(N_H = 4) - P_t(N_H = 4) \\ &= (P(4|4) - 1)P_t(4) + P(4|3)P_t(3) \end{aligned}$$

.

We expect, that after a long time, we reach equilibrium, where the probability distributions become static. Hence $\Delta P_\infty(4) = 0$. This gives us the equation

$$-\left(P(4|4) - 1\right) P_\infty(4) = P(4|3) P_\infty(3)$$
$$P_{\mathrm{SS}}(4) = \frac{1}{4} P_{\mathrm{SS}}(3).$$

We could do the same principle for all other macrostates. Inteterstingly, the probability to find the system in a macrostate with N_H heads is proportional to the amount of microstates that correspond to this macrostate.

This leads us to the FIRST ERGODIC POSTULATE for equilibrium statistical mechanics:

> For a closed system, every microstate which satisfies the global constraints is equally likely to be occupied. - Boltzmann

The second Ergodic postulate states:

> The macrostate which is occupied in thermal equilibrium is the one with the largest number of microstates.

Note that this macrostate must satisfy the fixed constraints (for example the total amount of coins).

One concept following from this is IRREVERSINILITY. If we have a ordered macrostate, i.e. one with only a single microstate, then the system will evolve to a disordered macrostate, i.e. one with many microstates.

4.1.1 Closed System

Consider a system of a fixed number of particles N and a fixed total energy E. The atoms cannot interact with the environment in any way. We want to count in how many ways we can distribute the energy E among the N particles. For this we make boxes for the energies ε_i and put n_i particles in each box. We have $\frac{N!}{n_1!(N-n_1)!}$ ways to put n_1 particles in the first box. For the second box we then get $\frac{(N-n_1)!}{n_2!(N-n_1-n_2)!}$ ways to put n_2 particles in the second box. Continuing this way, we get the total number of microstates

$$\Omega = \frac{N!}{n_1! n_2! \dots n_K!}.$$

Lec 10

We now want to maximize the number of states Ω, while being constrained by the total energy E and the number of particles N.

$$E = \sum_i n_i \varepsilon_i, \quad N = \sum_i n_i.$$

In statistical mechanics, it is typically easier to maximize $\ln \Omega$ instead of Ω. This is allowed since ln is a monotonic function. This is now a Lagrange Multiplier problem.

$$\frac{\partial}{\partial n_i}\left(\ln \Omega - \alpha\left(\sum_i n_i - N\right) - \beta\left(\sum_i n_i \varepsilon_i - E\right)\right) = 0.$$

We can focus on the first term

$$\frac{\partial}{\partial n_i} \ln \Omega = \frac{\partial}{\partial n_i}\left(\ln N! - \sum_j \ln n_j!\right).$$

Since we care about large N, we can use Stirling's approximation $\ln n! \simeq n \ln(n) - n$ to get

$$\frac{\partial}{\partial n_i}\left(\ln N! - \sum_j n_j \ln n_j + \sum_j n_j - \alpha\left(\sum_j n_j - N\right) - \beta\left(\sum_j n_j \varepsilon_j -\right.\right.$$
$$= -\ln n_i - 1 + 1 - \alpha - \beta\varepsilon_i = 0$$
$$\Rightarrow n_i = e^{-\alpha} e^{-\beta\varepsilon_i}.$$

Dividing by the total number of particles N gives us the probability to find a particle in the energy state ε_i:

$$p_i = \frac{n_i}{N} = \frac{e^{-\alpha} e^{-\beta\varepsilon_i}}{\sum_j e^{-\alpha} e^{-\beta\varepsilon_j}} = \frac{e^{-\beta\varepsilon_i}}{\sum_j e^{-\beta\varepsilon_j}}.$$

We define the SINGLE PARTICLE PARTITION FUNCTION as

$$Z_{\mathrm{sp}} = \sum_j e^{-\beta\varepsilon_j}.$$

4.2 System inside a thermal bath

Consider a system of N particles, in a microstate with energy E_s. We put this system inside a thermal bath with energy $E_b = E - E_s$. We also give the bath a temperature T. The system can exchange energy with the bath, but not particles.

In Thermodynamics, we have seen the first law of thermodynamics

$$\mathrm{d}E = \delta Q - \delta W.$$

Here E is the internal energy, δQ is the heat transferred to the system and δW is the work done by the system. For a system at temperature T, we have seen that

$$\sum_{\text{cycle}} \frac{\delta Q}{T} \geq 0.$$

The equality occurs for reversible processes. For a reversible process, we define

$$\mathrm{d}S = \frac{\delta Q}{T}.$$

Where $\mathrm{d}S$ is the change in entropy. For reversible processes, we can hence write the first law as

$$\mathrm{d}E = T\mathrm{d}S - P\mathrm{d}V.$$

From this we get

$$\left(\frac{\partial E}{\partial S}\right)_V = T, \quad \left(\frac{\partial S}{\partial E}\right)_V = \frac{1}{T}.$$

From this, we also find

$$\left(\frac{\partial^2 S}{\partial E^2}\right)_V = 0.$$

Returning to our bath, the number of states of the full system is given by

$$\Omega_{S+B} = \Omega_S(E_s)\Omega_B(E - E_s).$$

For now, we choose to look at one microstate of the system. Hence, $\Omega_S(E_s) = 1$. Thus

$$p_S(E_S) = A\Omega_B(E - E_s).$$

Boltzmann came up with a conjecture, that $S = k_B \ln \Omega$, where

$$k_B = 1.3806503 \times 10^{-23} \text{J/K}.$$

With this, we get

$$S(E-E_s) = k_B \ln \Omega_B(E-E_s) \Rightarrow \Omega_B(E-E_s) = e^{S(E-E_s)/k_B}.$$

Since $E_S \ll E$, we can do a Taylor expansion of $S(E-E_s)$ around E:

$$\begin{aligned} S(E-E_S) &= S(E) - E_S \left(\frac{\partial S}{\partial E}\right)_V + \frac{E_S^2}{2!}\left(\frac{\partial^2 S}{\partial E^2}\right)_V \\ &= S(E) - \frac{E_S}{T}. \end{aligned}$$

Where we assumed the temperature to be constant. Thus we get

$$\Omega_B(E-E_S) = e^{S(E)/k_B} e^{-E_S/(k_B T)}.$$

So we have

$$p_S = A e^{S(E)/k_B} e^{-E_S/(k_B T)}.$$

So once again, the probability is proportional to $e^{-\beta\varepsilon}$ with

$$\beta = \frac{1}{k_B T}.$$

Since the sum of the probabilities must be 1, we can find the normalization constant A:

$$p_s = \frac{e^{-E_S/(k_B T)}}{\sum_{E_S} e^{-E_S/(k_B T)}}.$$

We now define the partition function of the system as

$$Z = \sum_{E_S} e^{-E_S/(k_B T)}.$$

What we just derived is known as the Boltzmann distribution. It describes the probability that a *single* microstate of a system for which energy E_s is occupied when connected to a thermal bath at temperature T is

$$p_s = \frac{e^{-E_S/(k_B T)}}{Z}.$$

Where

$$Z = \sum_{E_S} e^{-E_S/(k_B T)}.$$

In many situations, we can say that $\Omega(E_S)$ states have energy E_S. The partition function then becomes

$$Z = \sum_{E_S} \Omega(E_S) e^{-E_S/(k_B T)}.$$

Typically $\Omega(E_S)$ is called the degeneracy of E_S. Notice that both definitions are the same. In the first we sum over the microstates, in the second we sum over the macrostates.

The question arises, how we can relate the "single particle" partition function to that of a system. If we have a system made of particles, we have

$$E_S = \varepsilon_1 + \varepsilon_2 + \ldots + \varepsilon_N.$$

We cam assume all particles are the same and independent. Hence,

$$Z = \sum_i e^{-E_i/k_B T} . \sum_j e^{-E_j/k_B T} \cdots = \sum \cdots \sum e^{-E_S/k_B T}.$$

We can also write this as

$$Z_{\text{system}} = Z_{\text{sp}}^N.$$

Where N is the number of particles in the system.

The partition function Z allows us to find any thermodynamic quantity. For example

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z,$$

where $\beta = 1/(k_B T)$. We can also find the entropy S:

$$S = k_B \ln Z + k_B T \frac{\partial Z}{\partial T}.$$

Or the free energy

$$F = -k_B T \ln Z.$$

For our system of particles, we have

$$Z_{\text{sys}} = Z_{\text{sp}}^N = \left(\sum_i e^{-\beta\varepsilon_i}\right)^N.$$

Now, the logarithm of the partition function is

$$\ln Z_{\text{sys}} = N \ln Z_{\text{sp}} = N \ln \left(\sum_i e^{-\beta\varepsilon_i}\right).$$

Differentiating with respect to β gives us

$$\frac{\partial}{\partial \beta} \ln Z_{\text{sys}} = N \frac{\sum_i -\varepsilon_i e^{-\beta\varepsilon_i}}{\sum_i e^{-\beta\varepsilon_i}} = -N\langle\varepsilon\rangle.$$

So we have

$$-\frac{\partial}{\partial \beta} \ln Z_{\text{sys}} = N\langle\varepsilon\rangle = \langle E \rangle.$$

Let us first look at quadratic energies. So imagine we have objects with energies

$$E(p_i, q_i) = a p_i^2 + b q_i^2.$$

We look at the phase space of a single particle. It consist of little boxes with width dp_i and dq_i. The probability to be in a state with energy $E(p_i, q_i)$ is given by

$$p(p_i, q_i) = \frac{e^{-\beta E(p_i, q_i)}}{\cdot} \frac{\mathrm{d}p_i \mathrm{d}q_i}{h}.$$

The constant ensures, we have probabilities. The unit of h is given by

$$\text{J s} = \text{kg m}^2/\text{s}.$$

Two comments: h will never affect mean energies etc. Secondly, h is NOT the Planck constant, although it has the same unit.

Since the single particle partition function is just a sum, we can instead use integrals to get

$$Z_{\text{sp}} = \int \mathrm{d}p_i \int \mathrm{d}q_i e^{-\beta E(p_i, q_i)}.$$

With the quadratic energy, we can write this as

$$\begin{aligned} Z_{\text{sp}} &= \int_0^\infty \mathrm{d}p_i \int_{\mathbb{R}} \mathrm{d}q_i e^{-\beta(ap_i^2+bq_i^2)} \\ &= \int_0^\infty \mathrm{d}p_i e^{-\beta a p_i^2} \cdot \int \mathrm{d}q_i e^{-\beta b q_i^2} \\ &= \sqrt{\frac{\pi}{\beta a}} \cdot \sqrt{\frac{\pi}{\beta b}} \cdot \frac{1}{2} \\ &= \frac{\pi}{2\beta\sqrt{ab}}. \end{aligned}$$

From this, we find the mean energy of $Z = Z_{\text{sp}}^N$:

$$\begin{aligned} \langle E \rangle &= -\frac{\partial}{\partial\beta} \ln Z = -N \frac{\partial}{\partial\beta} \ln Z_{\text{sp}} \\ &= -N \frac{\partial}{\partial\beta} \ln\left(\frac{\pi}{2\beta\sqrt{ab}}\right) = N\frac{1}{\beta} = Nk_BT. \end{aligned}$$

Looking at both the p_i and q_i contributions, we find that each contributes $\frac{1}{2}Nk_BT$. This is known as the EQUIPARTITION THEOREM. It states that each quadratic degree of freedom contributes $\frac{1}{2}k_BT$ to the mean energy.

In a free gas in 3D, $E(p_x, p_y, p_z) = \frac{p_x^2+p_y^2+p_z^2}{2m}$. Hence, we have 3 quadratic degrees of freedom. So for N particles, we have

$$\langle E \rangle = \frac{3}{2}Nk_BT.$$

Recall the beginning of statistical mechanics, we have

$$PV = Nm\langle v^2 \rangle = Nk_BT.$$

Here we used

$$\frac{3m\left\langle v^2 \right\rangle}{2} = \frac{3}{2}k_BT.$$

Lec 11

The equipartition theorem was very useful for atoms. However considering a molecule, in the simplest form, a diatomic molecule. This can be modelled with two point particles connected by a spring.

In this case, there are 6 degrees of freedom. 3 translational, 2 rotational and 1 vibrational. They will all be proportional to p_i^2. So from the motion, we expect the thermal energy to be

$$\langle E \rangle = \frac{6}{2}Nk_BT = 3Nk_BT.$$

Since the atoms are connected by a spring, we can also expect the potential energy to be proportional to x^2. Hence, we expect the thermal energy to be

$$\langle E \rangle = \frac{6}{2}Nk_BT + \frac{1}{2}Nk_BT = \frac{7}{2}Nk_BT.$$

The internal energy is not too easy to measure. Instead, one measures the heat capacity, which is defined by

$$C_V = \left(\frac{\partial \left\langle E \right\rangle}{\partial T}\right)_V.$$

In our case, we have

$$C_V = \frac{7}{2}Nk_B.$$

However, doing the measurement, we notice that there are 3 regimes. At low temperatures, the heat capacity is $C_V = \frac{3}{2}Nk_B$. At intermediate temperatures, the heat capacity is $C_V = \frac{5}{2}Nk_B$. At high temperatures, the heat capacity is $C_V = \frac{7}{2}Nk_B$.

The question now is, why do some degrees of freedom of molecules get effectively "frozen"?

The second scenario where the equipartition theorem fails is blackbody radiation. A body held at a fixed temperature emits radiation. We know that the spectrum of emission is identical, regardless of the surface used for the object. However, as we will see, the amount of radiation varies by material. We want to find a formula for the spectrum of radiation. This gives us the amount of energy we should assign to different frequencies of light.

Recall that light is an harmonic oscillator in the electromagnetic field. So the energy is given by

$$\varepsilon_0 \frac{E^2}{2} + \frac{B^2}{2\mu_0}.$$

So we have 2 quadratic degrees of freedom. Hence, we expect naively that we have an amount k_BT of energy / oscillator.

We consider EM waves inside a box, "oven". We consider shiny walls, metal, which seem simpler to work with. The wave equation reads

$$\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\mathbf{E} = \nabla^2\mathbf{E}.$$

We take the Ansatz, that the solution is of the form

$$\mathbf{E}(\mathbf{r}, t) = f(t)\mathbf{E}(\mathbf{r}).$$

Plugged into the wave equation, we find

$$\frac{1}{c^2}\frac{\mathrm{d}^2 f(t)}{\mathrm{d}t^2}\mathbf{E}(\mathbf{r}) = f(t)\nabla^2\mathbf{E}(\mathbf{r}).$$

Separating the variables, we find

$$\begin{aligned} \frac{1}{c^2}\frac{\mathrm{d}^2 f(t)}{\mathrm{d}t^2}\frac{1}{f(t)} &= \frac{\nabla^2\mathbf{E}(\mathbf{r})}{\mathbf{E}(\mathbf{r})} = -k^2 \\ \Rightarrow \frac{\mathrm{d}^2 f(t)}{\mathrm{d}t^2} &= -c^2k^2 f(t) \\ \Rightarrow \nabla^2\mathbf{E}(\mathbf{r}) &= -k^2\mathbf{E}(\mathbf{r}). \end{aligned}$$

Using $k^2 = k_x^2 + k_y^2 + k_z^2$ as a trial solution, we take

$$\mathbf{E}(\mathbf{r}) = \begin{pmatrix} E_x(\mathbf{r}) \\ E_y(\mathbf{r}) \\ E_z(\mathbf{r}) \end{pmatrix} = \begin{pmatrix} E_x \cos(k_x x)\sin(k_y y)\sin(k_z z) \\ E_y \sin(k_x x)\cos(k_y y)\sin(k_z z) \\ E_z \sin(k_x x)\sin(k_y y)\cos(k_z z) \end{pmatrix}.$$

The transverse fields will have E parallel to the wall that they hit. Hence, this must be zero if the walls are perfect conductors. Hence, since the E_x component comes from light propagating along z or y, it must be zero at the boundaries $z = 0$ and $z = L$. The first condition is fulfilled, since $\sin(0) = 0$. The second condition gives us

$$\sin(k_z L) = 0 \Rightarrow k_z = \frac{n_z\pi}{L}.$$

With the same argument for y and x, we find

$$k_x = \frac{n_x\pi}{L}, \quad k_y = \frac{n_y\pi}{L}, \quad k_z = \frac{n_z\pi}{L}.$$

We also have by Maxwell $\nabla \cdot E = 0$. This gives us the condition

$$k_x E_x + k_y E_y + k_z E_z = 0.$$

In summary, we get

$$k^2 = \frac{\pi^2}{L^2}(n_x^2 + n_y^2 + n_z^2).$$

There are more possible solutions, the higher n is. We want to do this explicitly. For this consider the space spanned by k_x, k_y, k_z. This is known as the K SPACE. This is useful, since we get a coordinate grid every $\frac{\pi}{L_i}$. Hence, the amount of states constraining a certain wavevector is given by the volume of a thin slice of a sphere. Since we require $n_x, n_y, n_z > 0$, we only take the first octant. Hence, we have that the "volume" of the slice is

$$\frac{1}{8} \cdot 4\pi k^2 \mathrm{d}k.$$

The volume occupied by a single state is given by

$$\frac{\pi^3}{L_x L_y L_z}.$$

Hence the number of solutions with k between k and $k+\mathrm{d}k$ is given by

$$\frac{1}{8} \cdot 4\pi k^2 \mathrm{d}k \cdot \frac{L_x L_y L_z}{\pi^3} = \frac{V k^2 \mathrm{d}k}{2\pi^2}.$$

Since we have 2 polarizations, we multiply by 2. Thus, we get

$$g(k)\mathrm{d}k = \frac{V k^2 \mathrm{d}k}{\pi^2}.$$

Where $g(k)$ is the degeneracy of the modes. Oftentimes its useful to write it in terms of frequencies

$$g(\nu)\mathrm{d}\nu = \frac{8\pi V \nu^2}{c^3}\mathrm{d}\nu.$$

Applying the equipartition theorem, we get that in thermal equilbrium, each mode (a harmonic oscillator) has an energy of $k_B T$. Hence, the energy density is given by

$$\rho(\nu)\mathrm{d}\nu = g(\nu) k_B T \mathrm{d}\nu = \frac{8\pi V \nu^2}{c^3} k_B T \mathrm{d}\nu.$$

This would mean that the world is full of gamma rays. This is known as the ULTRAVIOLET CATASTROPHE.

The solution to this was given by Planck. He proposed that the energy of light / harmonic oscillator comes in discrete units with size ε_i. For a single oscillator, we can write the partition function

$$Z = \sum_{n=0}^{\infty} e^{-\beta n \varepsilon_i} = \frac{1}{1 - e^{-\beta \varepsilon_i}}.$$

$$\begin{aligned}
\langle E \rangle &= -\frac{\partial}{\partial \beta} \ln Z_i = -\frac{\partial}{\partial \beta} \ln Z_i \\
&= -\frac{\partial}{\partial \beta} \ln\left(\frac{1}{1 - e^{-\beta \varepsilon_i}}\right) \\
&= -\frac{\partial}{\partial \beta}(-\ln(1 - e^{-\beta \varepsilon_i})) = -\frac{\partial}{\partial \beta}(-\ln(1 - e^{-\beta \varepsilon_i})).
\end{aligned}$$

Since $Z = \prod Z_i$, we have that

$$\langle E \rangle = \sum_i \frac{\varepsilon_i}{e^{\beta \varepsilon_i} - 1}.$$

Now, if $k_B T \gg \varepsilon_i$, then $\frac{\varepsilon_i}{k_B T} \ll 1$. Hence, we can expand the exponential and get

$$e^{\beta \varepsilon_i} \approx 1 + \beta \varepsilon_i.$$

Hence,

$$\langle E \rangle = \sum_i \frac{\varepsilon_i}{e^{\beta \varepsilon_i} - 1} = \sum_i \frac{\varepsilon_i}{\beta \varepsilon_i} = \sum_i k_B T.$$

Which is the classical result. However, if $k_B T \ll \varepsilon_i$, then

$$\langle E_i \rangle \approx \sum_i \varepsilon_i e^{-\beta \varepsilon_i}.$$

This is not the classical result. Intuitively, this means that high frequency modes can only take energies in steps of $h\nu_i$. If this is much bigger than the thermal energy, then the mode almost never gets excited.

To compute $\rho(\nu)\mathrm{d}\nu$ now, we have

$$\begin{aligned}
\rho(\nu)\mathrm{d}\nu &= \frac{1}{V} g(\nu)\mathrm{d}\nu \langle E \rangle \\
&= \frac{1}{V} g(\nu)\mathrm{d}\nu \frac{h\nu}{e^{\beta h\nu} - 1} \\
&= \frac{8\pi\nu^2}{c^3}\mathrm{d}\nu \frac{h\nu}{e^{\beta h\nu} - 1}.
\end{aligned}$$

This solves the UV problem, since if we integrate the spectral density, we do not get infinity. This distribution is known as the PLANCK DISTRIBUTION.

The maximum of this distribution is described by WIEN'S DISPLACEMENT LAW. For this, we set the derivative of the spectral density to zero. To make the math a bit easier, we define

$$x = \frac{h\nu}{k_B T}.$$

We can then write the spectral density as

$$\rho(\nu)\mathrm{d}\nu \propto T^3 x^3 \frac{1}{e^x - 1}\mathrm{d}\nu.$$

Doing a fixed point iteration, we find that

$$h\nu_{\max} = 2.821 k_B T.$$

Hence, $\nu_{\max} \propto T$. This is Wien's displacement law.

As a warning, the maximal frequency is not the same as the maximal wavelength. The maximal wavelength is given by

$$\frac{hc}{\lambda_{\max}} = 4.965 k_B T.$$

We now want to find the total energy. For this we use again the x from above and doing the integral, we get

$$U(T) = \int_0^\infty \rho(\nu)\mathrm{d}\nu = \frac{8\pi k_B^4 T^4}{c^3 h^3} \int_0^\infty \frac{x^3}{e^x - 1}\mathrm{d}x = \frac{8\pi k_B^4 T^4}{c^3 h^3} \frac{\pi^4}{15}.$$

The integral is not the important part here, but the fact that $U(T) = aT^4$. This is known as the STEFAN-BOLTZMANN LAW.

One question that arises, if it is ok to assume a box. Since the volume cancels out, this is indeed the case. So what about the material? For this, we can do a Gedankenexperiment. Consider two boxes, one with a perfect mirror and one with a perfect absorber. If we put them in thermal contact, then we have $T_1 \stackrel{!}{=} T_2 = T$. From the second law of thermodynamics

$$\int (\rho_{\text{shiny}} - \rho_{\text{black}}) \mathrm{d}\nu = 0.$$

In the transmission state, we now add a filter. This filter is so small that it only lets through a small range of frequencies $[\nu, \nu + \mathrm{d}\nu]$. Hence, we must have

$$\int_{\nu}^{\nu+\mathrm{d}\nu} (\rho_{\text{shiny}} - \rho_{\text{black}}) \mathrm{d}\nu = 0.$$

With this, we can conclude, that the spectral density is independent of the material. So what changes with the material? The walls couple differently. For this, we look at the flux emitted from a black body. Let j be the defined as the product of energy density, speed of light c and angular average. Hence

$$j = T^4\sigma = \frac{c}{4}U(T)..$$

Since emission and absorption are related by the second law of thermodynamics, we have that for a perfect black body the emission is σT^4, equal to the absorption. For a shiny body, the absorption is smaller, hence the emission is smaller. The law then is

$$j = e\sigma T^4.$$

Where e is the emissivity of the material ($0 \leq e \leq 1$).

We now want to look at the walls and how they emit. For this we zoom in on a atomic scale. If photons are emitted in quanta, they must also be absorbed in quanta. Hence, atoms also have to be quantized. The simplest model for this is an atom with two energy levels. Here, N_1 atoms are at the ground state E_1 and N_2 atoms are at the excited state E_2. Now, how do the energy transitions occur?

One option is (spontaneous emission). For this, an atom in E_2 emits at random. The probability to decay in the next $\mathrm{d}t$ is given by $A \cdot \mathrm{d}t$.

If you look at the emission of many atoms in E_2, then we will get an exponential decay of the population of the excited state. For this we get the differential equation

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = -AN_2 \Rightarrow N_2(t) = N_2(0)e^{-At}.$$

We define $\tau = \frac{1}{A}$ as the LIFETIME of the excited state. Thus,

$$\langle t \rangle = \int_0^\infty tAe^{-At}\mathrm{d}t = \frac{1}{A} = \tau.$$

This constant is atom + level specific.

The second option is absorption. Since light has to be there, we call this STIMULATED ABSORPTION. For this, the rate is proportional to $B_{1\to2}\rho(\nu)$. On average, it will thus take

$$\langle t \rangle = \frac{1}{B_{1\to2}\rho(\nu)}.$$

In equilibrium, we excpect that the number of atoms in E_1 and E_2 is constant. Hence, we have

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = -AN_2 + \rho(\nu)B_{1\to2}N_1.$$

Thus, $\rho(\nu) = \frac{AN_2}{B_{1\to2}N_1}$. We can also have STIMULATED EMISSION. For this, the rate is given by $B_{2\to1}\rho(\nu)$. Hence, in equilibrium, we have

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = -AN_2 - \rho(\nu)B_{2\to1}N_2 + \rho(\nu)B_{1\to2}N_1 = 0.$$

From this, we get

$$\rho(\nu) = \frac{AN_2}{B_{1\to2}N_1 - B_{2\to1}N_2} = \frac{A/B_{21}}{\frac{B_{12}}{B_{21}}\frac{N_1}{N_2} - 1}.$$

This is the right form for the Planck distribution. Hence, we have plugging it in

$$B_{21} = B_{12} = B \quad A = \frac{8\pi h\nu^3}{c^3}B.$$

These coefficients are called the EINSTEIN COEFFICIENTS.

So now, what about external states if ab atom, aka velocity? A moving atom emits at a Doppler shifted frequency,

$$\nu' = \nu(1 + \frac{v}{c}).$$

This influences the absorption aswell! We want to find the momentum distribution of the atoms. For this,

$$P(p)\mathrm{d}p = \frac{1}{Z}g(p)\mathrm{d}p \langle E \rangle .$$

The first thing we can do is draw the phase space in a grid, with coordinates q, p. We then have a region in phase space where the number of states is proportional to $\mathrm{d}^3p\mathrm{d}^3q$. This has a dimension of $[(Js)^3]$. We can set the number of states in a volume V to be $\frac{\mathrm{d}^3q\mathrm{d}^3p}{h^3}$. For now, h is just a unit. Thus, we get

$$g(p)\mathrm{d}p = \frac{1}{h^3}\int_V \mathrm{d}^3q \int_{\text{shell}} \mathrm{d}^3p = \frac{V}{h^3}4\pi p^2\mathrm{d}p.$$

Here, we once again used the k space shell. For a photon, with $p = hk$ we get

$$g(p)\mathrm{d}p = \frac{Vk^2}{2\pi^2}\mathrm{d}k.$$

Which is the same as before (with a factor of 2 for polarization). We define the Boltzmann weight

$$e^{-\beta E} = e^{-\beta\frac{p^2}{2m}}.$$

We lastly have to normalize the distribution. For this, we compute the partition function

$$Z = \int_0^\infty g(p)\mathrm{d}p e^{-p^2/2mk_BT} = \frac{V}{\lambda_{th}^3}.$$

Where λ_{th} is the THERMAL DE BROGLIE WAVELENGTH given by

$$\lambda_{th} = \frac{h}{\sqrt{2\pi mk_BT}}.$$

If we ask if this is classical, we have to compare λ_{th} to the average distance between atoms. If $\lambda_{th} \ll d$, for example with $N_2, 300K, 1$ bar, we get

$$\lambda_{th} = 0.019 \text{ nm} \ll d = 3.4 \text{ nm}.$$

However, for electrons in copper at room temperature, this does not work.

Putting everything together, we get the MAXWELL-BOLTZMANN DISTRIBUTION

$$P(p)\mathrm{d}p = \frac{4\pi p^2}{(2\pi m k_B T)^{3/2}} e^{-p^2/2mk_BT}\mathrm{d}p.$$

This is essentially a Gaussian distribution multiplied by a p^2 factor.

5 Wave-Particle Duality

Lec 13

So far, we've seen that waves can extend in space and can diffract and interfere. For a plane wave, the simplest way to describe them is as

$$e^{i(kx-\omega t)}.$$

Particles on the other hand, are localised at a single point in (x, p). One famous experiment is the DAVISSON-GERMER experiment. For this, a source of electrons is fired at a Nickel crystal. Then the intensity at different angles is measured.

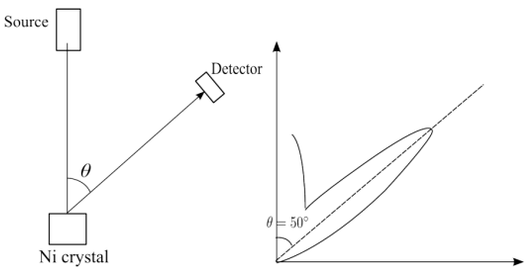


Figure 13: The Davisson-Germer experiment.

They excpected to see the results for a particle. I.e. there is specular reflection (i.e. surface is smooth) and there is diffusive reflection (i.e. surface is rough). Since the crystal is not a perfect mirror, they expected to see a distribution proportional to $\cos\theta$. However, what was instead measured was a diffraction pattern.

This observation was resolved by the DE BROGLIE HYPOTHESIS, postulated 2 years prior. It is an analogue to Planck's hypothesis, which gave photons a momentum $p = \frac{h}{\lambda}$. De Broglie proposed that all matter also has a wavelength associated with it, given by

$$\lambda = \frac{h}{p}.$$

For non-relativistic particles, this gives

$$E_{\text{kin}} = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}.$$

So why do we not see diffraction patterns for matter? The answer lies, that if we put a beam of particles onto a screen with an aperture on order of a wavelength, we will get the intensity pattern

$$I(\theta) = I(0)\text{sinc}^2(\pi \sin\theta \Delta_x/\lambda).$$

With this, we find the first minimum at

$$\sin\theta_{\text{min}} = \frac{\lambda}{\Delta_x}.$$

Thus, we find that if $\lambda \ll \Delta_x$, then we get particle-like behaviour. If $\lambda \sim \Delta_x$, then we get wave-like behaviour.

For example, an electron with 1eV of kinetic energy has a wavelength of 1.2nm. This can usually not be seen since diffraction gratings are on the order of 100nm.

Having wave lengths this short, can be used in electron microscopy. It consists of a high voltage electric gun, some magnetic lenses and a fluorescent screen. In the best case,

the resolution can be as good as $\Delta x = \frac{\lambda}{NA} \propto \lambda$. This it is useful for small things. However, the sample has to be thin and robust, since we are shooting electrons at it. This is a problem for biological samples, since they are not robust.

However instead of using a transmission electron microscope (TEM), we can use a scanning electron microscope (SEM). For both TEM and SEM, the limit of realistic resolution is about 0.5nm. This is mainly because of aberrations.

Other limitations to the diffraction limit include lens aberrations, energy spread of the electron beam. We furthermore are limited to not be able to see something smaller than the wavelength of the electron. This is where Neutron scattering comes in. For this, several things have to change. First, the source becomes a nuclear reactor, since neutrons are not charged. Second, the lenses are not possible, since neutrons are not charged. Lastly, the detector cannot be fluorescent, since neutrons are not charged. Instead, we do more nuclear reactions, which emit charged particles, which can then be detected.

We want to resolve $\lambda \approx 1\text{Å}$. We find $\lambda = \frac{h}{\sqrt{2m_n E_{\text{kin}}}}$. For this, we need $E_{\text{kin}} \approx 0.1\text{eV}$. In nuclear reactions, we typically have energies around 1MeV. Thus, we need to slow down the neutrons. This is done by using a moderator, which is a material with light nuclei. The neutrons scatter off the light nuclei and lose energy.

5.1 Localised particles as wave packets

We still have to describe why the plane wave is everywhere and why we detect particles in one point. The idea is that the particle does not originate from open space. Instead, we imagine a plane wave that is localised in space. For this, we use a wave function

$$\psi(x,t) = \int_{k_0-\Delta k}^{k_0+\Delta k} A(k) e^{i(kx-\omega t)} dk.$$

Writing $E_{\text{kin}} = \hbar\omega$ and $p = \hbar k$, we can write this as

$$\omega(k) = \frac{\hbar k^2}{2m}.$$

This gives us a dispersion relation. In optics $\omega \propto k$, which is a linear dispersion relation. This is not the case for matter waves, since $\omega \propto k^2$. We can expand $\omega(k)$ as

$$\omega(k) = \omega_0 + \frac{\hbar k_0}{m}\kappa + \frac{\hbar}{2m}\kappa^2,$$

where $\kappa = k - k_0$. For small κ, we can neglect the last term. This gives us

$$\omega(k) = \omega_0 + v_g \kappa,$$

where v_g is a group velocity. Thus we get

$$e^{i(kx-\omega t)} = e^{i(k_0 x - \omega_0 t)} e^{i\kappa(x - v_g t)}.$$

Integrating the wave function over κ gives us

$$\Re\psi(x,t) \propto \frac{\sin(\Delta k(x - \hbar k_0 t/m))}{x - \hbar k_0 t/m} \cdot \cos(k_0 x - \omega_0 t).$$

The sinc function is the envelope of the wave packet, which moves with a group velocity $v_g = \hbar k_0/m$. The cosine gives us a fast oscillation. The first zero, which gives us the width of the wave packet, is given by

$$\Delta x = \frac{2\pi}{\Delta k} \Leftrightarrow \Delta x \Delta k = 2\pi.$$

Plugging in $p = \hbar k$, we get

$$\Delta x \Delta p = h.$$

This is not Heisenberg's uncertainty principle, but it is close to it. With this we can also determine how fast the wave moves. For this we look at the maximum of the wave packet. I.e. when the argument of the sinc function is zero. This gives us

$$x_{\max}(t) = \frac{\hbar k_0}{m} t.$$

Differentiating this gives us the velocity of the wave packet, which is

$$v_{\max} = \frac{\hbar k_0}{m} = \frac{p}{m} = v_g.$$

The magnitude of the wave packet can tell us about the probability of finding a particle at a certain position. This is given by

$$P(x,t) = |\psi(x,t)|^2.$$

If we now look at the diffraction as a postion measurement. If p is sharp, then we have no information on x. Behind the slit we know x up to Δx. Also $\theta_1 = \frac{\Delta p_x}{p}$.

6 Quantum Mechanics

Two important people, when it comes to Quantum Mechanics are Schrödinger, who came up with a wave equation for a function called "the wavefunction", and Heisenberg, who came up with matrix mechanics.

Lets recall, that in classical mechanics, our theory describes the physical system we are interested in. For example given a collection of particles, we can describe them with positions and momenta. Furthermore the theory gives us rules for how the system changes in time. In classical mechanics, this is given by Newtons laws.

In Quantum Mechanics, our theory must provide three things. It must provide a description of a system, the wavefunction. It must describe the evolution, the Schrödinger equation. Lastly we have to describe how to extract information, measurement in Quantum mechanics.

6.1 The wavefunction

Every physical system is completely specified by a complex-valued function known as the WAVEFUNCTION. We denote the wavefunction as

$$\psi(x,t).$$

The wavefunction produces a probability density by

$$\psi^*(x,t)\psi(x,t)\mathrm{d}x = |\psi(x,t)|^2\mathrm{d}x.$$

Since this should be a probability, it must be normalized, so

$$\int_{-\infty}^{\infty} |\psi(x,t)|^2\mathrm{d}x = 1. \tag{6.1}$$

As a bad example, consider the free particle. It has a known momentum $p = \hbar k$ and $E = \hbar\omega$. We have the wavefunction

$$\psi(x,t) = Ae^{i(kx-\omega t)} = Ae^{i(px-Et)/\hbar}.$$

This is not strictly normalizable, which is why this is a bad example.

Since the wavefunction tells us everything about the particle. Since ψ only depends on x, t, there must be a mathematically equivalent description in terms of other properties.

We thus want to introduce a function $\tilde{\psi}(p,t)$, which is the wavefunction in momentum space. How do we do this? We will use orthonormal expansions. Since we will use them a lot, we will quickly go over them. As a first example, consider the Fourier series. Let there be an even real function specified on $[-\pi,\pi]$. Then we can expand it into a cosine series

$$\psi(x) = \sum_n \tilde{\psi}_n \cos(nx). \tag{6.2}$$

We know, that the cosine functions are orthogonal and normalized, so orthonormal, i.e.

$$\int_{-\pi}^{\pi} \cos(nx)\cos(mx)\mathrm{d}x = \pi\delta_{n,m}.$$

So integrating eq. (6.2) multiplied by $\cos(mx)$ gives

$$\tilde{\psi}_m = \frac{1}{\pi}\int_{-\pi}^{\pi} \psi(x)\cos(mx)\mathrm{d}x.$$

These $\tilde{\psi}_m$ give a full description of the function $\psi(x)$. Be warned, that we just considered even functions on the interval $[-\pi,\pi]$. This allowed the use of cosine functions. To fully describe any function, we need a complete set of orthonormal functions. In this case, this would be cosine and sine functions.

For a general complex function (for example wavefunction), we can express them in terms of a complete set of orthonormal functions $\phi_n(x)$, as

$$\psi(x) = \sum_n \tilde{\psi}_n\phi_n(x). \tag{6.3}$$

We require that

$$\int \phi_n^*(x)\phi_m(x)\mathrm{d}x = \delta_{n,m}.$$

Analogous to the Fourier series, we can find the coefficients $\tilde{\psi}_n$ by

$$\tilde{\psi}_m = \int \phi_m^*(x)\psi(x)\mathrm{d}x.$$

Hence, the $\tilde{\psi}_m$ give a full description of the function $\psi(x)$. But we may worry that we have a flaw, similar to the Cosine Series, i.e. if we can go back to the original function. Substituting our result into eq. (6.3) gives

$$\psi(x) = \sum_n \int \phi_n^*(x')\psi(x')\mathrm{d}x'\phi_n(x).$$

Switching the order of summation and integration gives

$$\psi(x) = \int \mathrm{d}x'\psi(x')\sum_n \phi_n^*(x')\phi_n(x).$$

If the second half is the Dirac delta function $\delta(x-x')$, then we have the desired result.

Definition 6.1: Dirac delta function

The Dirac delta function is defined as

$$\delta(x) = \int_{-\infty}^{\infty} e^{2\pi i\varepsilon x}\mathrm{d}\varepsilon.$$

This notion is only really well defined inside an integral. The important relation is

$$\int_{-\infty}^{\infty} f(x)\delta(x)\mathrm{d}x = f(0).$$

This also yields

$$\int_{-\infty}^{\infty} \delta(x)\mathrm{d}x = 1.$$

A useful relation to remember is

$$\delta(ax) = \frac{1}{|a|}\delta(x).$$

Instead of a discrete set of orthonormal functions, we can also have a continuous set. They are defined in a very similar way.

$$\psi(x) = \int \tilde{\psi}(\alpha)\phi(\alpha,x)\mathrm{d}\alpha.$$

Again, we require orthonormality, i.e.

$$\int \phi^*(\alpha, x)\phi(\beta, x)\mathrm{d}x = \delta(\alpha - \beta).$$

Doing the same trick as before, but using Fubini's theorem, we find

$$\tilde{\psi}(\beta) = \int \phi^*(\beta, x)\psi(x)\mathrm{d}x.$$

We also require completeness, i.e.

$$\int \phi^*(\alpha, x')\phi(\alpha, x)\mathrm{d}\alpha = \delta(x - x').$$

6.2 Physical Variables as Operators

Let us look at a free particle again, with momentum $p = \hbar k$ and energy $E = \hbar\omega$. The wavefunction is

$$\psi(x, t) = Ae^{i(kx-\omega t)} = Ae^{i(px-Et)/\hbar}.$$

Playing around with this wavefunction, we find that applying the operator $\frac{\partial}{\partial x}$ gives

$$\frac{\partial}{\partial x}\psi(x, t) = ik\psi(x, t) = \frac{ip}{\hbar}\psi(x, t).$$

Similarly, applying the operator $\frac{\partial}{\partial t}$ gives

$$\frac{\partial}{\partial t}\psi(x, t) = -i\omega\psi(x, t) = -\frac{iE}{\hbar}\psi(x, t).$$

This motivates a postulate of Quantum Mechanics:

Each physical quantity has an associated operator. Furthermore, if you measure a physical quantity, you will obtain an eigenvalue of the operator associated with the physical quantity.

For example, from the above, we can see that the momentum operator is

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}. \tag{6.4}$$

And for the energy operator, we have

$$\hat{E} = i\hbar\frac{\partial}{\partial t}. \tag{6.5}$$

Since these operators correspond to physical quantities, the Eigenvalues must be real!

Let's try to express a general wavefunction $\psi(x, t)$ in terms of momentum. For this, we first need an orthonormal set of functions, which we find as the Eigenfunctions of the momentum operator.

$$\hat{p}\phi(p, x) = p\phi(p, x).$$

Performing the operation as in eq. (6.4) gives

$$-i\hbar\frac{\partial}{\partial x}\phi(p, x) = p\phi(p, x) \Rightarrow \phi(p, x) = C(p)e^{ipx/\hbar}.$$

These are the eigenfunction of the momentum operator. We want to verify they are indeed orthonormal. We have

$$\begin{aligned}\int \phi^*(p, x)\phi(p', x)\mathrm{d}x &= \int C^*(p)e^{-ipx/\hbar}C(p')e^{ip'x/\hbar}\mathrm{d}x \\ &= C^*(p)C(p')\int e^{i(p'-p)x/\hbar}\mathrm{d}x \\ &= C^*(p)C(p')2\pi\hbar\delta(p' - p).\end{aligned}$$

To satisfy orthonormality, we require

$$C^*(p)C(p')2\pi\hbar\delta(p' - p) = \delta(p' - p) \Rightarrow C(p) = \frac{1}{\sqrt{2\pi\hbar}}.$$

Thus

$$\phi(p, x) = \frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}. \tag{6.6}$$

Thus,

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}}\int e^{ipx/\hbar}\tilde{\psi}(p)\mathrm{d}p.$$

This is the Fourier transform, and $\tilde{\psi}(p)$ is the wavefunction in momentum space. Doing things the other way around, we find

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}}\int e^{-ipx/\hbar}\psi(x)\mathrm{d}x.$$

This momentum wavefunction, must also be a complete description. It also gives a new probability density,

$$\tilde{\psi}^*(p, t)\tilde{\psi}(p, t)\mathrm{d}p = |\tilde{\psi}(p, t)|^2\mathrm{d}p.$$

One example of superposition would be that taking two waves with oposite momenta,

$$\psi(x) = \frac{1}{\sqrt{2}}\left(\frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}} + \frac{e^{-ipx/\hbar}}{\sqrt{2\pi\hbar}}\right) = \frac{1}{\sqrt{2}}(\phi(p, x)+\phi(-p, x)).$$

So here we don't have to do the integral explicitly, since we saw two eigenfunctions of the momentum operator. This is a superposition of two momentum eigenstates, with momenta p and $-p$.

Index

Boltzmann distribution, 16
Bragg scattering, 12
Brewster angle, 7

circular polarisation, 6
Compton scattering, 12
Compton wavelength, 12

Davisson-Germer, 20
de Broglie hypothesis, 20
degeneracy, 16
dynodes, 11

Einstein coefficients, 19
envelope, 9
equipartition theorem, 17
expectation value, 14

Fabry-Perot etalon, 9
filament, 11
Finesse, 10
first Ergodic postulate, 15
focal length, 2
focal plane, 2
focal point, 2
Fraunhofer approximation, 4
Free spectral range, 10
free-space wavelength, 1
Full width half maximum, 10

grating, 7
Grating Spectrometer, 7

Huygen's principle, 4

irreversinility, 15

k space, 18

lens equation, 2
lifetime, 19
linear polarisation, 6

magnification, 2
Maxwell-Boltzmann distribution, 20
Michelson Spectrometer, 8

optical path, 1

p-polarisation, 6
parallele Polarisation, 6
paraxial approximation, 2
photon, 11
phototube, 11
Planck constant, 11
Planck distribution, 18
Planck experiment, 11

quanta, 11

Rayleigh criterion, 5

s-polarisation, 6
senkrechte Polarisation, 6
single particle partition function, 15
Stefan-Boltzmann law, 18
stimulated absorption, 19
stimulated emission, 19

thermal de Broglie wavelength, 19
thermionic emission, 11

ultraviolet catastrophe, 18

wavefunction, 22
Wien's displacement law, 18
work function, 11